

Управление парком кластеров

Подходы, проблемы, решения

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Задача

- Команда выбрала модель **множества кластеров** — и столкнулась с тем, что «поднять» их — это ещё полбеды
- Нужен ответ не на «Какой один инструмент крутой?», а на то, **как устроить процесс**: жизненный цикл, обновления, единообразие, контуры (DMZ, CDE, etc)



План

- ✓ От архитектора ждут **сравнимые варианты**: открытый стек, готовая платформа, корпоративный продукт — с понятными компромиссами
- ✓ Сегодня разберём **несколько подходов** и после каждого зафиксируем выводы — что получаем и чем платим





Cluster API (Quick Start, Proof Of Concept)



Introduction - The Cluster API B X

cluster-api.sigs.k8s.io

Private browsing

Sign in

Introduction

Quick Start

Quick Start Operator

Concepts

Manifesto

1. Tasks

1.1. Certificate Management

1.1.1. Using Custom Certificates

1.1.2. Generating a Kubeconfig

1.1.3. Auto Rotate Certificates in KCP

1.2. Bootstrap

1.2.1. Kubeadm based bootstrap

1.2.1.1. Kubelet configuration

1.2.2. MicroK8s based bootstrap

1.3. Upgrading management and workload clusters

1.4. External etcd

1.5. Using kustomize

1.6. Upgrading Cluster API

The Cluster API Book

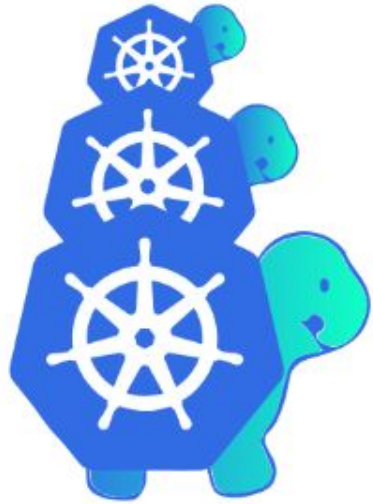
Kubernetes Cluster API

Cluster API is a Kubernetes sub-project focused on providing declarative APIs and tooling to simplify provisioning, upgrading, and operating multiple Kubernetes clusters.

Started by the Kubernetes Special Interest Group (SIG) [Cluster Lifecycle](#), the Cluster API project uses Kubernetes-style APIs and patterns to automate cluster lifecycle management for platform operators. The supporting infrastructure, like virtual machines, networks, load balancers, and VPCs, as well as the Kubernetes cluster configuration are all defined in the same way that application developers operate deploying and managing their workloads. This enables consistent and repeatable cluster deployments across a wide variety of infrastructure environments.

Getting started

- [Quick Start](#)
- [Concepts](#)
- [Developer guide](#)
- [Contributing](#)
- [Videos explaining Cluster API architecture](#)



Concepts - The Cluster API Book

cluster-api.sigs.k8s.io/user/concepts

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1.4. External etcd

1.5. Using kustomize

1.6. Upgrading Cluster API

The Cluster API Book

Concepts

Similar to how you can use StatefulSets or Deployments in Kubernetes to manage a group of Pods, in Cluster API you can use custom resources like KubeadmControlPlane (a control plane implementation) to manage a set of control plane Machines, or you can use MachineDeployments to manage a group of worker Machines, each one of them representing a host server and the corresponding Kubernetes Node.

Extensibility is at the core of Cluster API and Cluster API providers like Cluster API provider VSphere, AWS, GCP etc. can be used to deploy Cluster API managed Clusters to your preferred infrastructure, as well as to configure many other parts of the system.

Management Cluster
Kubernetes

Namespace

Cluster API
<<Controller>>

Custom Resources

Cluster

InfrastructureCluster

ControlPlanes

Machine

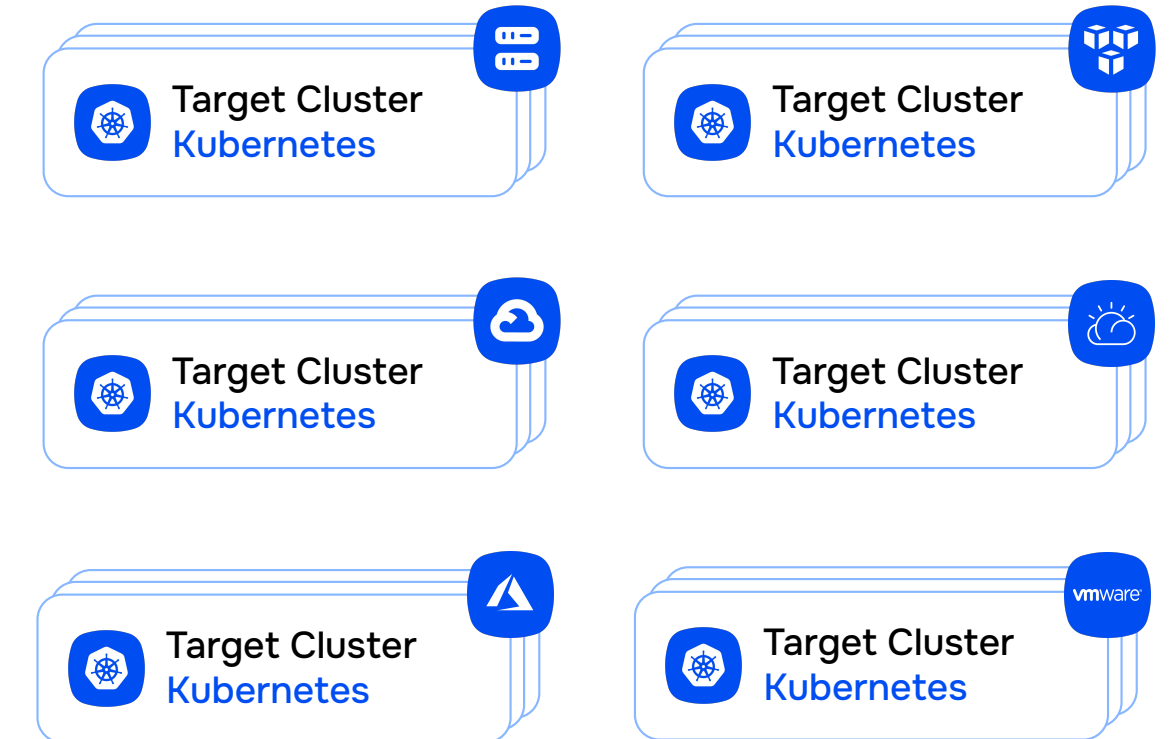
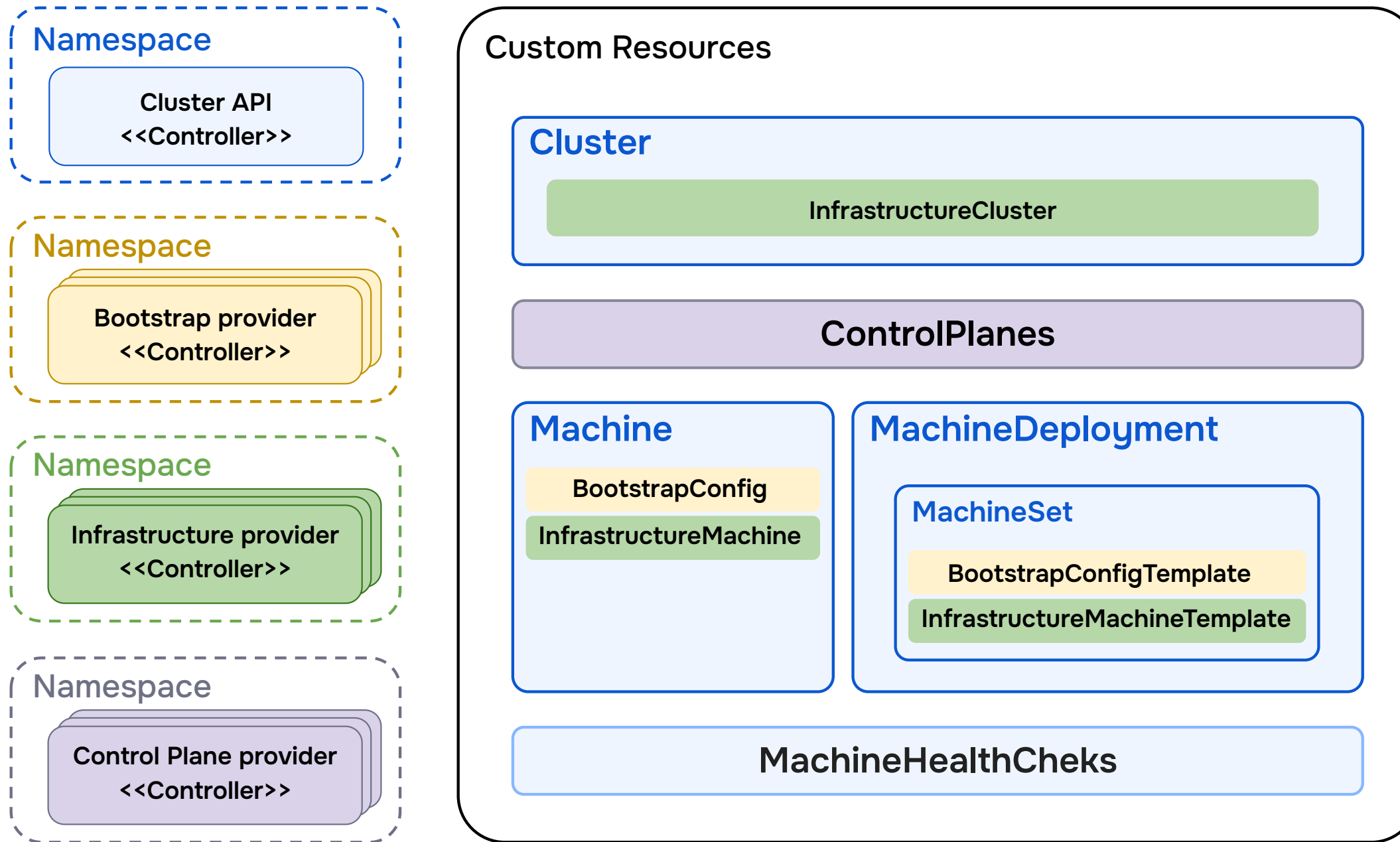
MachineDeployment

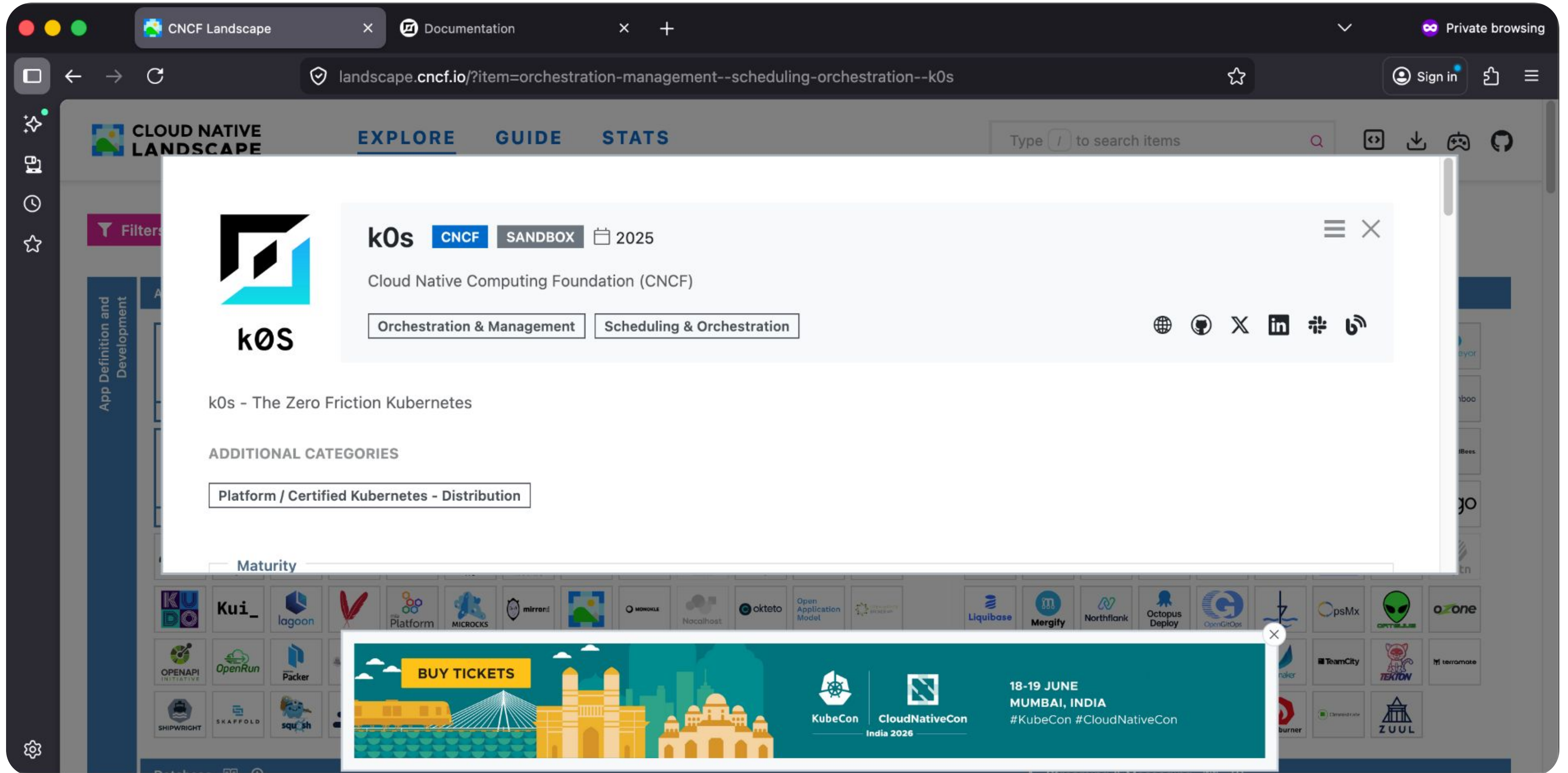
MachineSet

Namespace

Bootstrap provider
<<Controller>>

Management Cluster Kubernetes





<https://landscape.cncf.io/?item=orchestration-management--scheduling-orchestration--k0s>

CNCF Landscape

k0s | Kubernetes distribution for X

Private browsing

k0sproject.io

Sign in

k0sZero Friction Kubernetes

DocsBlogSupport

The Simple, Solid & Certified Kubernetes Distribution

THE KUBERNETES FOR EDGE

(Pronounced "K-zero-ess")

Deploy and run Kubernetes workloads at any scale on any infrastructure. CNCF certified. 100% open source & free.

GET STARTED

k0s is the simple, solid & certified Kubernetes distribution that works on any infrastructure: bare-metal, on-premises, edge, IoT, public & private clouds. It's 100% open source & free.

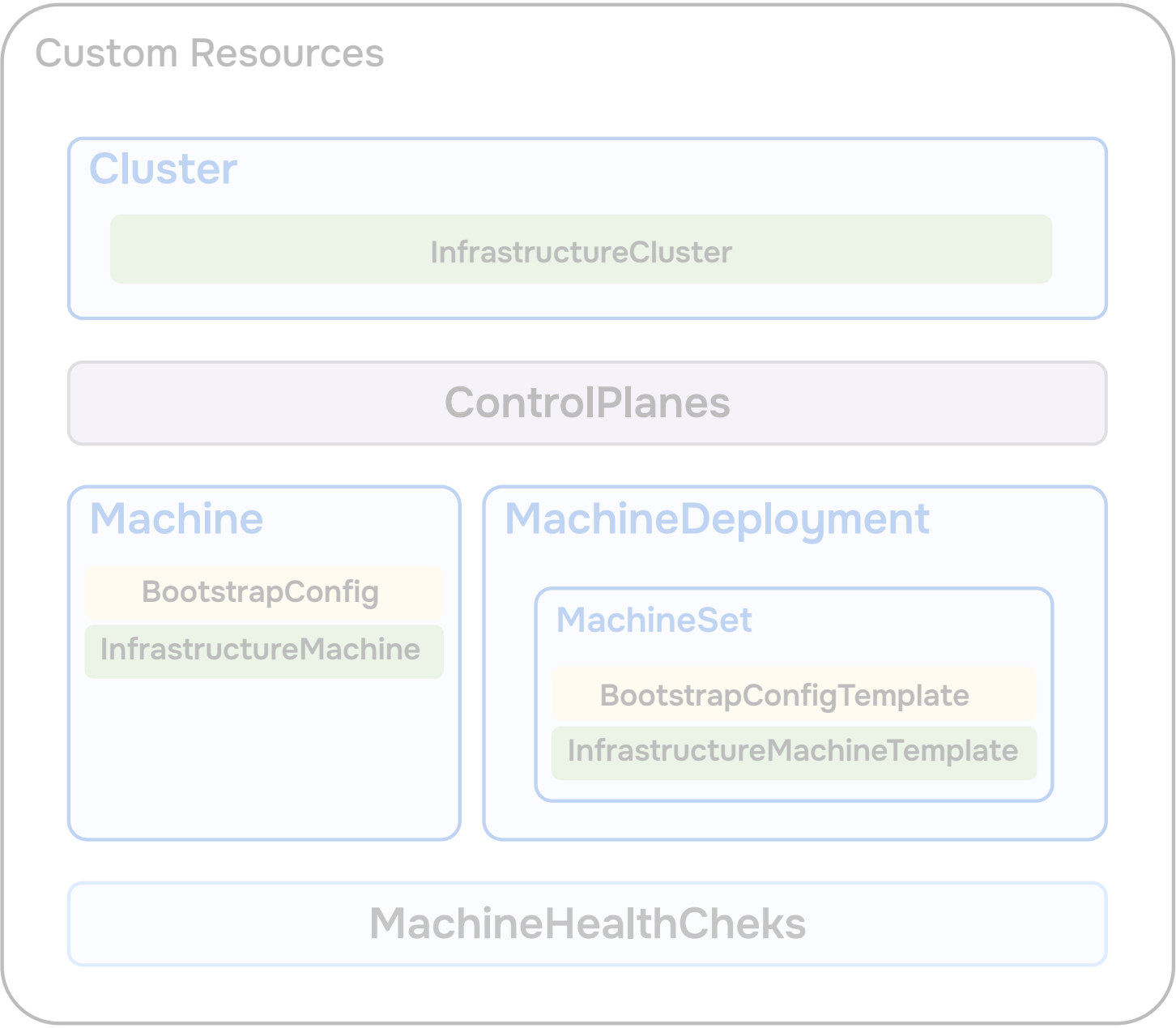
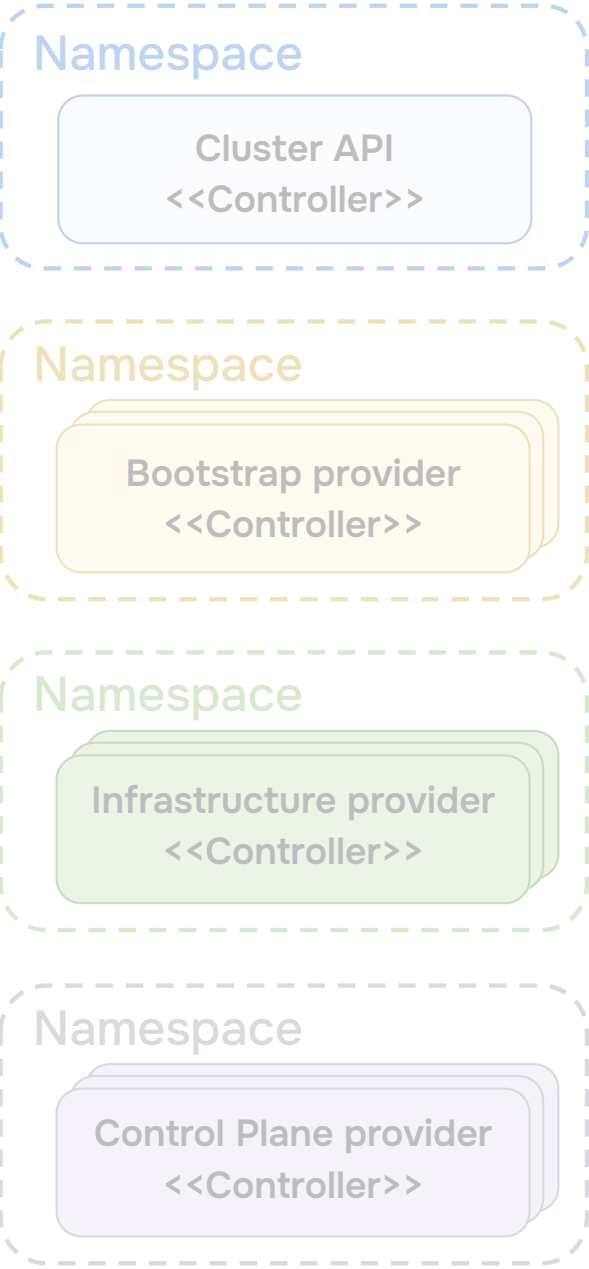
```
$ curl -sSf https://get.k0s.sh | sudo sh
$ sudo k0s install controller --single
$ sudo k0s start # wait a minute
$ sudo k0s kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
k0s	Ready	<none>	4m6s	v1.35.1+k0s

<https://k0sproject.io/>


```
[kovalkov@archmgmt:~$ curl -sSf https://get.k0s.sh | sudo sh
Downloading k0s from URL: https://github.com/k0sproject/k0s/releases/download/v1.35.2+k0s.0/k0s-v1.35.2+k0s.0-amd64
k0s is now executable in /usr/local/bin
You can use it to complete the installation of k0s on this node,
see https://docs.k0sproject.io/stable/install/ for more information.
[kovalkov@archmgmt:~$ sudo k0s install controller --single
[kovalkov@archmgmt:~$ sudo k0s start # wait a minute
[kovalkov@archmgmt:~$ sleep 60 && sudo k0s kubectl get nodes
NAME          STATUS    ROLES          AGE    VERSION
archmgmt      Ready     control-plane   51s    v1.35.2+k0s
[kovalkov@archmgmt:~$ sudo k0s kubectl get pods --all-namespaces
NAMESPACE     NAME                                READY   STATUS    RESTARTS   AGE
kube-system   coredns-5cc4b78c7c-lghfx           1/1     Running   0           75s
kube-system   kube-proxy-vvcc9                   1/1     Running   0           71s
kube-system   kube-router-7jcdh                   1/1     Running   0           71s
kube-system   metrics-server-df68c566c-m2fw9     1/1     Running   0           70s
kovalkov@archmgmt:~$
```


Management Cluster Kubernetes




```
[kovalkov@archmgmt:~$ sudo k0s kubeconfig admin > .kube/config]
```

```
[kovalkov@archmgmt:~$ clusterctl init]
```

Fetching providers

Installing cert-manager version="v1.19.4"

Waiting for cert-manager to be available...

spec.privateKey.rotationPolicy: In cert-manager >= v1.18.0, the default value changed from `Never` to `Always`.

Installing provider="cluster-api" version="v1.12.4" targetNamespace="capi-system"

spec.privateKey.rotationPolicy: In cert-manager >= v1.18.0, the default value changed from `Never` to `Always`.

Installing provider="bootstrap-kubeadm" version="v1.12.4" targetNamespace="capi-kubeadm-bootstrap-system"

spec.privateKey.rotationPolicy: In cert-manager >= v1.18.0, the default value changed from `Never` to `Always`.

Installing provider="control-plane-kubeadm" version="v1.12.4" targetNamespace="capi-kubeadm-control-plane-system"

spec.privateKey.rotationPolicy: In cert-manager >= v1.18.0, the default value changed from `Never` to `Always`.

Your management cluster has been initialized successfully!

You can now create your first workload cluster by running the following:

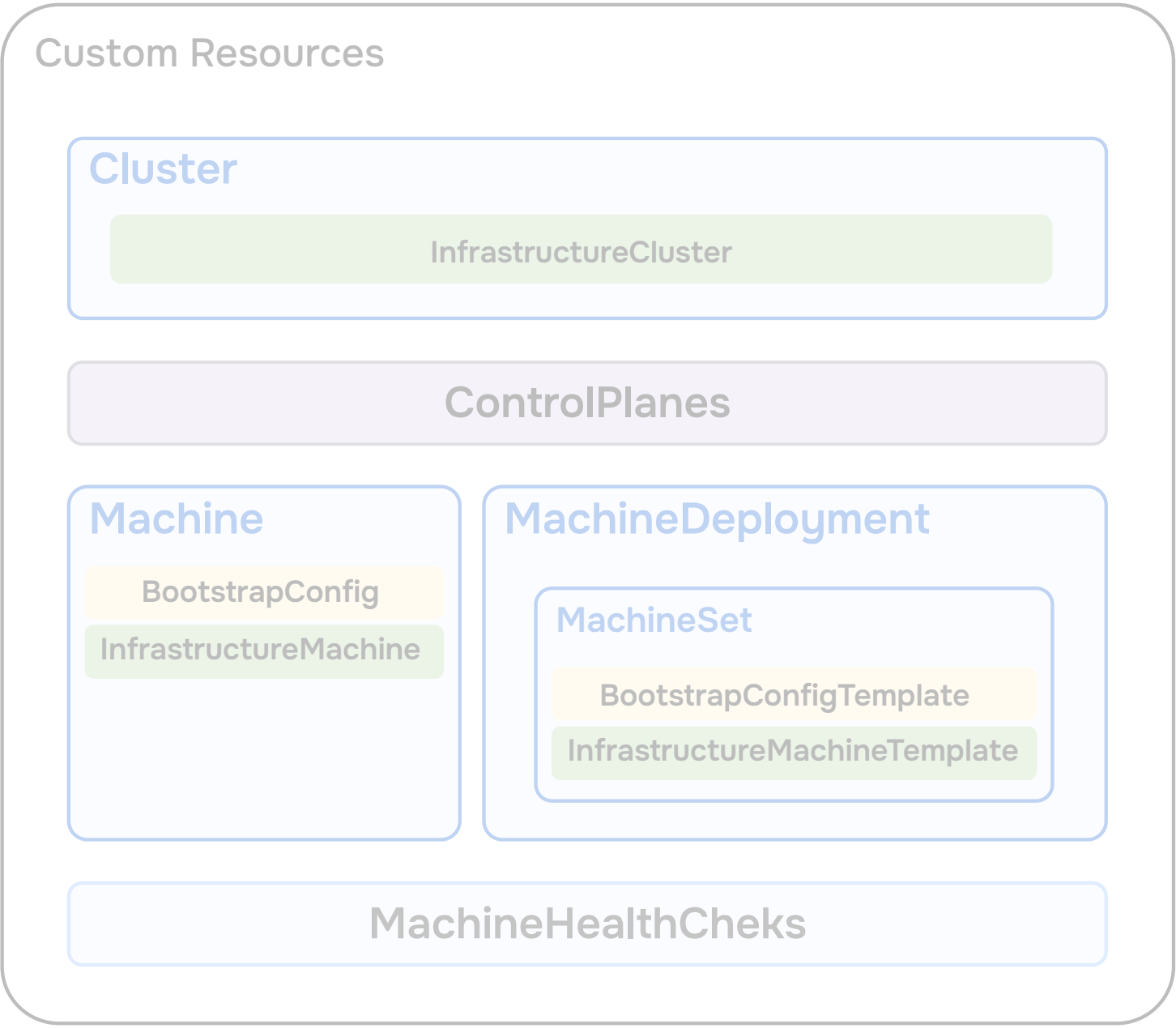
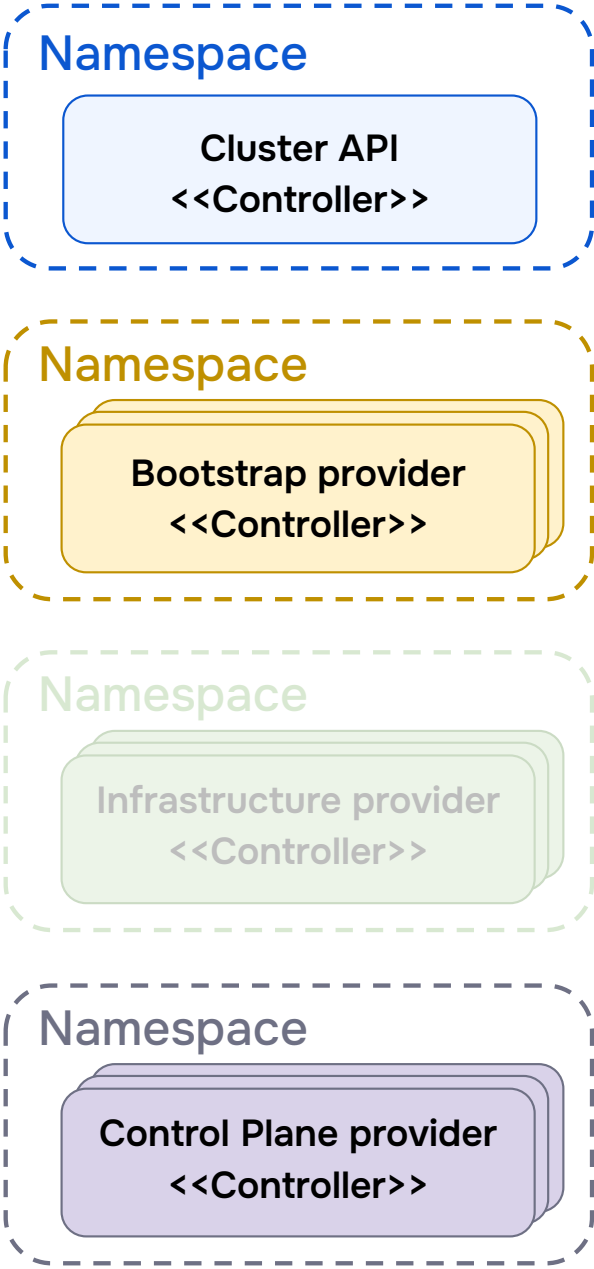
```
clusterctl generate cluster [name] --kubernetes-version [version] | kubectl apply -f -
```

```
[kovalkov@archmgmt:~$ kubectl get pods -A]
```

NAMESPACE	NAME	READY	STATUS	RESTARTS	AGE
capi-kubeadm-bootstrap-system	capi-kubeadm-bootstrap-controller-manager-74d59d776b-fpkv6	1/1	Running	0	52s
capi-kubeadm-control-plane-system	capi-kubeadm-control-plane-controller-manager-548cf6cf4d-rbghj	1/1	Running	0	49s
capi-system	capi-controller-manager-7c9b6c5bd9-n7qfk	1/1	Running	0	53s
cert-manager	cert-manager-87564587f-zk8xc	1/1	Running	0	71s
cert-manager	cert-manager-cainjector-679f58f748-5qjs7	1/1	Running	0	71s
cert-manager	cert-manager-webhook-6b69dc77cf-lm2rr	1/1	Running	0	71s
kube-system	coredns-5cc4b78c7c-lghfx	1/1	Running	0	20m
kube-system	kube-proxy-vvcc9	1/1	Running	0	20m
kube-system	kube-router-7jcdh	1/1	Running	0	20m
kube-system	metrics-server-df68c566c-m2fw9	1/1	Running	0	20m

```
kovalkov@archmgmt:~$
```

Management Cluster Kubernetes



Quick Start - The Cluster API BoXCNCF LandscapeXk0s | Kubernetes distribution for XGitHub - yandex-cloud/cluster-eX+Private browsing

github.com/yandex-cloud/cluster-api-provider-yandexSign in

READMEContributingApache-2.0 license

Провайдер Yandex Cloud для Kubernetes® Cluster API

`cluster-api-provider-yandex` — провайдер для развертывания кластера Kubernetes в облачной инфраструктуре [Yandex Cloud](#) с помощью [Kubernetes Cluster API](#).

Кластер разворачивается на базе [виртуальных машин](#) Yandex Compute Cloud и [L7-балансировщика](#) Yandex Application Load Balancer.

Преимущества создания кластера с помощью провайдера Yandex Cloud

- интеграция с API Yandex Cloud;
- декларативный подход к созданию и управлению кластером;
- кластер как [CustomResourceDefinition](#);
- широкий спектр параметров для конфигурации вычислительных ресурсов кластера;
- [пользовательские образы](#) операционных систем для мастера и узлов;
- пользовательский Control Plane;
- альтернатива [Terraform](#) в CI-процессах.

Совместимость провайдера с Cluster API

Версия провайдера	Версия Cluster API	Совместимость
-------------------	--------------------	---------------

<https://github.com/yandex-cloud/cluster-api-provider-yandex>


```
[kovalkov@archmgmt:~$ git clone https://github.com/yandex-cloud/cluster-api-provider-yandex.git
```

```
Cloning into 'cluster-api-provider-yandex'...
```

```
remote: Enumerating objects: 764, done.
```

```
remote: Counting objects: 100% (214/214), done.
```

```
remote: Compressing objects: 100% (114/114), done.
```

```
remote: Total 764 (delta 130), reused 101 (delta 100), pack-reused 550 (from 2)
```

```
Receiving objects: 100% (764/764), 276.29 KiB | 1.96 MiB/s, done.
```

```
Resolving deltas: 100% (374/374), done.
```

```
[kovalkov@archmgmt:~$ cd cluster-api-provider-yandex/
```

```
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ export IMG=cr.yandex/crpsjg1coh47p81vh2lc/capy/cluster-api-provider-yandex:latest
```

```
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ make install
```

```
mkdir -p /home/kovalkov/cluster-api-provider-yandex/bin
```

```
GOBIN=/home/kovalkov/cluster-api-provider-yandex/bin go install sigs.k8s.io/controller-tools/cmd/controller-gen@v0.16.5
```

```
go: downloading sigs.k8s.io/controller-tools v0.16.5
```

```
go: downloading github.com/spf13/cobra v1.8.1
```

```
go: downloading github.com/gobuffalo/flect v1.0.3
```

```
go: downloading k8s.io/apiextensions-apiserver v0.31.2
```

```
go: downloading k8s.io/apimachinery v0.31.2
```

```
go: downloading k8s.io/utils v0.0.0-20240711033017-18e509b52bc8
```

```
go: downloading golang.org/x/tools v0.26.0
```

```
go: downloading gopkg.in/yaml.v2 v2.4.0
```

```
go: downloading github.com/fatih/color v1.18.0
```

```
go: downloading k8s.io/api v0.31.2
```

```
go: downloading gopkg.in/yaml.v3 v3.0.1
```

```
go: downloading sigs.k8s.io/yaml v1.4.0
```

```
go: downloading github.com/spf13/pflag v1.0.5
```

```
go: downloading github.com/gogo/protobuf v1.3.2
```

```
go: downloading github.com/google/gofuzz v1.2.0
```

```
go: downloading k8s.io/klog/v2 v2.130.1
```

```
go: downloading sigs.k8s.io/structured-merge-diff/v4 v4.4.1
```

```
go: downloading github.com/mattn/go-colorable v0.1.13
```

```
go: downloading github.com/mattn/go-isatty v0.0.20
```

```
go: downloading golang.org/x/sync v0.8.0
```

```
go: downloading sigs.k8s.io/json v0.0.0-20221116044647-bc3834ca7abd
```



```
go: downloading github.com/modern-go/concurrent v0.0.0-20180306012644-bacd9c7ef1dd
go: downloading github.com/modern-go/reflect2 v1.0.2
go: downloading github.com/x448/float16 v0.8.4
go: downloading golang.org/x/text v0.19.0
/home/kovalkov/cluster-api-provider-yandex/bin/controller-gen rbac:roleName=manager-role crd webhook paths="./..." output:crd:artifact
s:config=config/crd/bases
GOBIN=/home/kovalkov/cluster-api-provider-yandex/bin go install sigs.k8s.io/kustomize/kustomize/v5@v5.3.0
go: downloading sigs.k8s.io/kustomize/kustomize/v5 v5.3.0
go: downloading github.com/spf13/cobra v1.7.0
go: downloading sigs.k8s.io/kustomize/api v0.16.0
go: downloading sigs.k8s.io/kustomize/cmd/config v0.13.0
go: downloading sigs.k8s.io/kustomize/kyaml v0.16.0
go: downloading github.com/go-errors/errors v1.4.2
go: downloading golang.org/x/text v0.13.0
go: downloading golang.org/x/exp v0.0.0-20231006140011-7918f672742d
go: downloading github.com/davecgh/go-spew v1.1.1
go: downloading k8s.io/kube-openapi v0.0.0-20230601164746-7562a1006961
go: downloading github.com/monochromegane/go-gitignore v0.0.0-20200626010858-205db1a8cc00
go: downloading github.com/xlab/treeprint v1.2.0
go: downloading gopkg.in/evanphx/json-patch.v5 v5.6.0
go: downloading google.golang.org/protobuf v1.30.0
go: downloading github.com/google/go-cmp v0.5.9
go: downloading github.com/google/shlex v0.0.0-20191202100458-e7afc7fbc510
go: downloading go.starlark.net v0.0.0-20200306205701-8dd3e2ee1dd5
/home/kovalkov/cluster-api-provider-yandex/bin/kustomize build config/crd | kubectl apply -f -
# Warning: 'commonLabels' is deprecated. Please use 'labels' instead. Run 'kustomize edit fix' to update your Kustomization automatica
lly.
# Warning: 'patchesStrategicMerge' is deprecated. Please use 'patches' instead. Run 'kustomize edit fix' to update your Kustomization
automatically.
customresourcedefinition.apiextensions.k8s.io/yandexclusters.infrastructure.cluster.x-k8s.io created
customresourcedefinition.apiextensions.k8s.io/yandexidentities.infrastructure.cluster.x-k8s.io created
customresourcedefinition.apiextensions.k8s.io/yandexmachines.infrastructure.cluster.x-k8s.io created
customresourcedefinition.apiextensions.k8s.io/yandexmachinetemplates.infrastructure.cluster.x-k8s.io created
kovalkov@archmgmt:~/cluster-api-provider-yandex$
```




```
kovalkov@archmgmt: ~/cluster-api-provider-yandex — ssh kovalkov@84.252.130.206 — 134x34
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ vim /home/kovalkov/yc-sa-key
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl create namespace capy-system
namespace/capy-system created
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl --namespace capy-system create secret generic yc-sa-key --from-file=key=/home/kovalkov/yc-sa-key
secret/yc-sa-key created
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ make deploy
GOBIN=/home/kovalkov/cluster-api-provider-yandex/bin go install sigs.k8s.io/controller-tools/cmd/controller-gen@v0.16.5
/home/kovalkov/cluster-api-provider-yandex/bin/controller-gen rbac:roleName=manager-role crd webhook paths="./..." output:crd:artifacts:config=config/crd/bases
cd config/manager && /home/kovalkov/cluster-api-provider-yandex/bin/kustomize edit set image controller=cr.yandex/crpsjg1coh47p81vh2lc/capy/cluster-api-provider-yandex:latest
/home/kovalkov/cluster-api-provider-yandex/bin/kustomize build bases | kubectl apply -f -
# Warning: 'vars' is deprecated. Please use 'replacements' instead. [EXPERIMENTAL] Run 'kustomize edit fix' to update your Kustomization automatically.
# Warning: 'commonLabels' is deprecated. Please use 'labels' instead. Run 'kustomize edit fix' to update your Kustomization automatically.
# Warning: 'patchesStrategicMerge' is deprecated. Please use 'patches' instead. Run 'kustomize edit fix' to update your Kustomization automatically.
Warning: resource namespaces/capy-system is missing the kubectl.kubernetes.io/last-applied-configuration annotation which is required by kubectl apply. kubectl apply should only be used on resources created declaratively by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.
namespace/capy-system configured
customresourcedefinition.apiextensions.k8s.io/yandexclusters.infrastructure.cluster.x-k8s.io configured
customresourcedefinition.apiextensions.k8s.io/yandexidentities.infrastructure.cluster.x-k8s.io unchanged
customresourcedefinition.apiextensions.k8s.io/yandexmachines.infrastructure.cluster.x-k8s.io unchanged
customresourcedefinition.apiextensions.k8s.io/yandexmachinetemplates.infrastructure.cluster.x-k8s.io unchanged
serviceaccount/capy-controller-manager created
role.rbac.authorization.k8s.io/capy-leader-election-role created
clusterrole.rbac.authorization.k8s.io/capy-manager-role created
clusterrole.rbac.authorization.k8s.io/capy-metrics-reader created
clusterrole.rbac.authorization.k8s.io/capy-proxy-role created
rolebinding.rbac.authorization.k8s.io/capy-leader-election-rolebinding created
clusterrolebinding.rbac.authorization.k8s.io/capy-manager-rolebinding created
```



```
kovalkov@archmgmt: ~/cluster-api-provider-yandex — ssh kovalkov@84.252.130.206 — 134x34
cd config/manager && /home/kovalkov/cluster-api-provider-yandex/bin/kustomize edit set image controller=cr.yandex/crpsjg1coh47p81vh2lc
/capy/cluster-api-provider-yandex:latest
/home/kovalkov/cluster-api-provider-yandex/bin/kustomize build bases | kubectl apply -f -
# Warning: 'vars' is deprecated. Please use 'replacements' instead. [EXPERIMENTAL] Run 'kustomize edit fix' to update your Kustomizati
on automatically.
# Warning: 'commonLabels' is deprecated. Please use 'labels' instead. Run 'kustomize edit fix' to update your Kustomization automatica
lly.
# Warning: 'patchesStrategicMerge' is deprecated. Please use 'patches' instead. Run 'kustomize edit fix' to update your Kustomization
automatically.
Warning: resource namespaces/capy-system is missing the kubectl.kubernetes.io/last-applied-configuration annotation which is required
by kubectl apply. kubectl apply should only be used on resources created declaratively by either kubectl create --save-config or kubec
tl apply. The missing annotation will be patched automatically.
namespace/capy-system configured
customresourcedefinition.apiextensions.k8s.io/yandexclusters.infrastructure.cluster.x-k8s.io configured
customresourcedefinition.apiextensions.k8s.io/yandexidentities.infrastructure.cluster.x-k8s.io unchanged
customresourcedefinition.apiextensions.k8s.io/yandexmachines.infrastructure.cluster.x-k8s.io unchanged
customresourcedefinition.apiextensions.k8s.io/yandexmachinetemplates.infrastructure.cluster.x-k8s.io unchanged
serviceaccount/capy-controller-manager created
role.rbac.authorization.k8s.io/capy-leader-election-role created
clusterrole.rbac.authorization.k8s.io/capy-manager-role created
clusterrole.rbac.authorization.k8s.io/capy-metrics-reader created
clusterrole.rbac.authorization.k8s.io/capy-proxy-role created
rolebinding.rbac.authorization.k8s.io/capy-leader-election-rolebinding created
clusterrolebinding.rbac.authorization.k8s.io/capy-manager-rolebinding created
clusterrolebinding.rbac.authorization.k8s.io/capy-proxy-rolebinding created
service/capy-controller-manager-metrics-service created
service/capy-webhook-service created
deployment.apps/capy-controller-manager created
Warning: spec.privateKey.rotationPolicy: In cert-manager >= v1.18.0, the default value changed from `Never` to `Always`.
certificate.cert-manager.io/capy-serving-cert created
issuer.cert-manager.io/capy-selfsigned-issuer created
mutatingwebhookconfiguration.admissionregistration.k8s.io/capy-mutating-webhook-configuration created
validatingwebhookconfiguration.admissionregistration.k8s.io/capy-validating-webhook-configuration created
kovalkov@archmgmt:~/cluster-api-provider-yandex$
```



```

kovalkov@archmgmt: ~/cluster-api-provider-yandex — ssh kovalkov@84.252.130.206 — 134x34

[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl -n capy-system get pods
NAME                                READY   STATUS              RESTARTS   AGE
capy-controller-manager-67698dd977-dr7zs  1/2     ImagePullBackOff    0          95s
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl -n capy-system describe pods capy-controller-manager-67698dd977-dr7zs | tail -n 10
Normal    Scheduled    2m7s                default-scheduler  Successfully assigned capy-system/capy-controller-manager-67698dd977-dr7zs to archmgmt
Normal    Pulling      2m7s                kubelet            spec.containers{manager}: Pulling image "cr.yandex/crpsjg1coh47p81vh2lc/capy/cluster-api-provider-yandex:latest"
Normal    Pulled       2m5s                kubelet            spec.containers{manager}: Successfully pulled image "cr.yandex/crpsjg1coh47p81vh2lc/capy/cluster-api-provider-yandex:latest" in 2.078s (2.078s including waiting). Image size: 37205342 bytes.
Normal    Created      2m5s                kubelet            spec.containers{manager}: Container created
Normal    Started      2m5s                kubelet            spec.containers{manager}: Container started
Normal    Pulling      43s (x4 over 2m5s)  kubelet            spec.containers{kube-rbac-proxy}: Pulling image "gcr.io/kubebuilder/kube-rbac-proxy:v0.13.1"
Warning   Failed       43s (x4 over 2m5s)  kubelet            spec.containers{kube-rbac-proxy}: Failed to pull image "gcr.io/kubebuilder/kube-rbac-proxy:v0.13.1": rpc error: code = NotFound desc = failed to pull and unpack image "gcr.io/kubebuilder/kube-rbac-proxy:v0.13.1": failed to resolve reference "gcr.io/kubebuilder/kube-rbac-proxy:v0.13.1": gcr.io/kubebuilder/kube-rbac-proxy:v0.13.1: not found
Warning   Failed       43s (x4 over 2m5s)  kubelet            spec.containers{kube-rbac-proxy}: Error: ErrImagePull
Normal    BackOff      2s (x8 over 2m5s)  kubelet            spec.containers{kube-rbac-proxy}: Back-off pulling image "gcr.io/kubebuilder/kube-rbac-proxy:v0.13.1"
Warning   Failed       2s (x8 over 2m5s)  kubelet            spec.containers{kube-rbac-proxy}: Error: ImagePullBackOff
kovalkov@archmgmt:~/cluster-api-provider-yandex$

```


Quick Start - The Cluster API Bo

CNCF Landscape

GitHub - yandex-cloud/cluster-e

kube-rbac-proxy unavailable · Is

Private browsing

github.com/yandex-cloud/cluster-api-provider-yandex/issues/48

Sign in

code

Issues

Pull requests

Actions

Projects

Security and quality

Insights

kube-rbac-proxy unavailable #48

New issue

Open

unreturned opened 4 days ago

...

Line should be updated to registry.k8s.io/kubebuilder/kube-rbac-proxy

cluster-api-provider-yandex/config/default/manager_auth_proxy_patch.yaml

Line 34 in 65c3d9e

34 image: gcr.io/kubebuilder/kube-rbac-proxy:v0.13.1

by [smallstep/step-issuer#335](#) (comment)

unreturned 4 days ago

Author ...

[kubernetes-sigs/cluster-api-addon-provider-helm#318](#)

Assignees

No one assigned

Labels

No labels

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

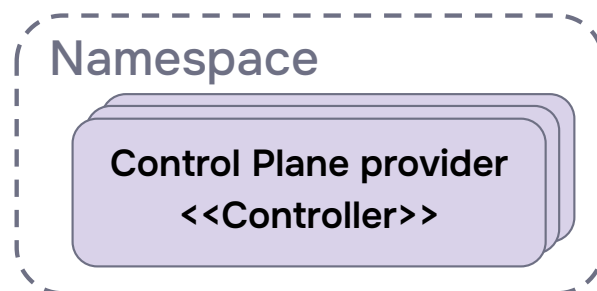
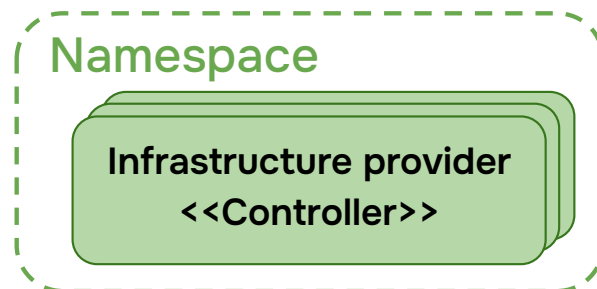
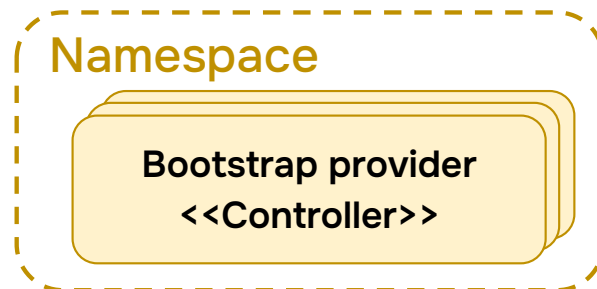
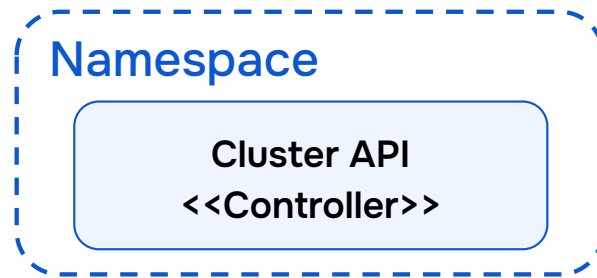
None yet

https://github.com/yandex-cloud/cluster-api-provider-yandex/issues/48


```
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl -n capy-system get pods
NAME                                READY    STATUS              RESTARTS   AGE
capy-controller-manager-67698dd977-dr7zs  1/2     ImagePullBackOff    0          95s
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl -n capy-system describe pods capy-controller-manager-67698dd977-dr7zs | tail -n 10
Normal    Scheduled    2m7s                default-scheduler    Successfully assigned capy-system/capy-controller-manager-67698dd977-dr7zs to archmgmt
Normal    Pulling     2m7s                kubelet              spec.containers{manager}: Pulling image "cr.yandex/crpsjg1coh47p81vh21c/capy/cluster-api-provider-yandex:latest"
Normal    Pulled      2m5s                kubelet              spec.containers{manager}: Successfully pulled image "cr.yandex/crpsjg1coh47p81vh21c/capy/cluster-api-provider-yandex:latest" in 2.078s (2.078s including waiting). Image size: 37205342 bytes.
Normal    Created     2m5s                kubelet              spec.containers{manager}: Container created
Normal    Started     2m5s                kubelet              spec.containers{manager}: Container started
Normal    Pulling     43s (x4 over 2m5s)  kubelet              spec.containers{kube-rbac-proxy}: Pulling image "gcr.io/kubebuilder/kube-rbac-proxy:v0.13.1"
Warning   Failed      43s (x4 over 2m5s)  kubelet              spec.containers{kube-rbac-proxy}: Failed to pull image "gcr.io/kubebuilder/kube-rbac-proxy:v0.13.1": rpc error: code = NotFound desc = failed to pull and unpack image "gcr.io/kubebuilder/kube-rbac-proxy:v0.13.1": failed to resolve reference "gcr.io/kubebuilder/kube-rbac-proxy:v0.13.1": gcr.io/kubebuilder/kube-rbac-proxy:v0.13.1: not found
Warning   Failed      43s (x4 over 2m5s)  kubelet              spec.containers{kube-rbac-proxy}: Error: ErrImagePull
Normal    BackOff     2s (x8 over 2m5s)  kubelet              spec.containers{kube-rbac-proxy}: Back-off pulling image "gcr.io/kubebuilder/kube-rbac-proxy:v0.13.1"
Warning   Failed      2s (x8 over 2m5s)  kubelet              spec.containers{kube-rbac-proxy}: Error: ImagePullBackOff
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl -n capy-system edit deployments.apps capy-controller-manager
deployment.apps/capy-controller-manager edited
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl -n capy-system get pods -w
NAME                                READY    STATUS              RESTARTS   AGE
capy-controller-manager-7c9df464bd-bfgpn  2/2     Running            0          17s
^C[kovalkov@archmgmt:~/cluster-api-provider-yandex$
```



Management Cluster Kubernetes



Custom Resources

Cluster

InfrastructureCluster

ControlPlanes

Machine

BootstrapConfig

InfrastructureMachine

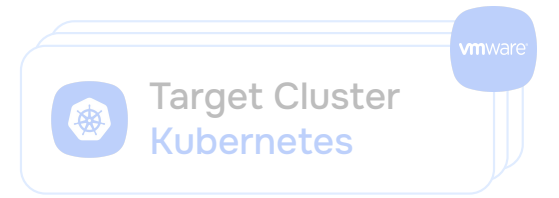
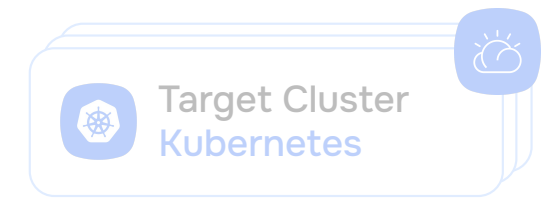
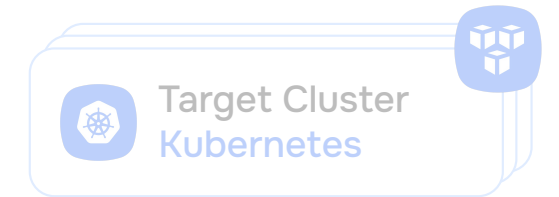
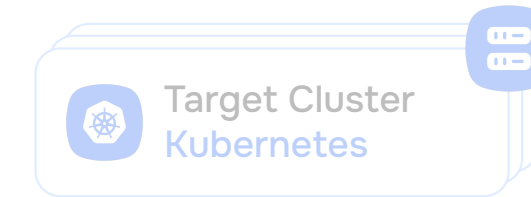
MachineDeployment

MachineSet

BootstrapConfigTemplate

InfrastructureMachineTemplate

MachineHealthChecks




```
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl get crd | grep cluster.x-k8s.io
clusterclasses.cluster.x-k8s.io                2026-04-04T22:39:57Z
clusterresourcesetbindings.addons.cluster.x-k8s.io 2026-04-04T22:39:57Z
clusterresourcesets.addons.cluster.x-k8s.io       2026-04-04T22:39:57Z
clusters.cluster.x-k8s.io                      2026-04-04T22:39:57Z
extensionconfigs.runtime.cluster.x-k8s.io        2026-04-04T22:39:58Z
ipaddressclaims.ipam.cluster.x-k8s.io           2026-04-04T22:39:58Z
ipaddresses.ipam.cluster.x-k8s.io               2026-04-04T22:39:58Z
kubeadmconfigs.bootstrap.cluster.x-k8s.io        2026-04-04T22:40:00Z
kubeadmconfigtemplates.bootstrap.cluster.x-k8s.io 2026-04-04T22:40:00Z
kubeadmcontrolplanes.controlplane.cluster.x-k8s.io 2026-04-04T22:40:02Z
kubeadmcontrolplanetemplates.controlplane.cluster.x-k8s.io 2026-04-04T22:40:02Z
machinedeployments.cluster.x-k8s.io             2026-04-04T22:39:58Z
machinedrainrules.cluster.x-k8s.io              2026-04-04T22:39:58Z
machinehealthchecks.cluster.x-k8s.io            2026-04-04T22:39:58Z
machinepools.cluster.x-k8s.io                   2026-04-04T22:39:58Z
machines.cluster.x-k8s.io                       2026-04-04T22:39:59Z
machinesets.cluster.x-k8s.io                    2026-04-04T22:39:59Z
providers.clusterctl.cluster.x-k8s.io           2026-04-04T22:39:36Z
yandexclusters.infrastructure.cluster.x-k8s.io   2026-04-04T22:50:11Z
yandexidentities.infrastructure.cluster.x-k8s.io 2026-04-04T22:50:11Z
yandexmachines.infrastructure.cluster.x-k8s.io   2026-04-04T22:50:11Z
yandexmachinetemplates.infrastructure.cluster.x-k8s.io 2026-04-04T22:50:11Z
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ ]
```


Quick Start - The Cluster API Bo ×CNCF Landscape ×GitHub - yandex-cloud/cluster-e ×kube-rbac-proxy unavailable · Is ×+Private browsing

github.com/yandex-cloud/cluster-api-provider-yandexSign in

READMEContributingApache-2.0 license

- о [каталог](#);
- о [сеть](#);
- о [подсеть](#) в выбранной зоне доступности.

3. Передайте идентификаторы ресурсов в переменные окружения:

```
export YANDEX_CONTROL_PLANE_MACHINE_IMAGE_ID=<идентификатор_образа>
export YANDEX_FOLDER_ID=<идентификатор_каталога>
export YANDEX_NETWORK_ID=<идентификатор_сети>
export YANDEX_SUBNET_ID=<идентификатор_подсети>
export YANDEX_ZONE_ID=<идентификатор_зоны_доступности>
```

4. Сформируйте манифесты кластера:

```
clusterctl generate cluster <имя_создаваемого_кластера> \
--from templates/cluster-template.yaml > /tmp/capy-cluster.yaml
```

По умолчанию согласно сгенерированному манифесту, для доступа в кластер будет развернут [L7-балансировщик](#) Application Load Balancer с динамическим внутренним IP-адресом. Вы можете [присвоить L7-балансировщику фиксированный IP-адрес](#).

Important


```
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ vim ya_env
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ source ya_env
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ clusterctl generate cluster archtest --worker-machine-count 1 --from templates/cluste
r-template.yaml > archtest-cluster.yaml
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ tail -n 20 archtest-cluster.yaml
---
apiVersion: bootstrap.cluster.x-k8s.io/v1beta1
kind: KubeadmConfigTemplate
metadata:
  name: archtest-worker
  namespace: default
spec:
  template:
    spec:
      joinConfiguration:
        nodeRegistration:
          kubeletExtraArgs:
            cloud-provider: external
          name: '{{ v1.local_hostname }}'
      preKubeadmCommands:
      - hostname "{{ v1.local_hostname }}"
      - echo "::1          ipv6-localhost ipv6-loopback" >/etc/hosts
      - echo "127.0.0.1    localhost" >>/etc/hosts
      - echo "127.0.0.1    {{ v1.local_hostname }}" >>/etc/hosts
      - echo "{{ v1.local_hostname }}" >/etc/hostname
kovalkov@archmgmt:~/cluster-api-provider-yandex$
```



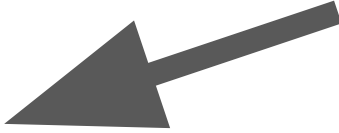
```
apiVersion: cluster.x-k8s.io/v1beta1
kind: Cluster
metadata:
  name: archtest
  namespace: default
spec:
  clusterNetwork:
    pods:
      cidrBlocks:
        - 172.25.0.0/16
    services:
      cidrBlocks:
        - 172.26.0.0/16
  controlPlaneRef:
    apiVersion: controlplane.cluster.x-k8s.io/v1beta1
    kind: KubeadmControlPlane
    name: archtest-control-plane
  infrastructureRef:
    apiVersion: infrastructure.cluster.x-k8s.io/v1alpha1
    kind: YandexCluster
    name: archtest
```



```
apiVersion: controlplane.cluster.x-k8s.io/v1beta1
kind: KubeadmControlPlane
metadata:
  name: archtest-control-plane
  namespace: default
spec:
  kubeadmConfigSpec:
    initConfiguration:
      localAPIEndpoint:
        bindPort: 8443
      nodeRegistration:
        kubeletExtraArgs:
          cloud-provider: external
        name: '{{ v1.local_hostname }}'
    joinConfiguration:
      controlPlane:
        localAPIEndpoint:
          bindPort: 8443
      nodeRegistration:
        kubeletExtraArgs:
          cloud-provider: external
        name: '{{ v1.local_hostname }}'
  machineTemplate:
    infrastructureRef:
      apiVersion: infrastructure.cluster.x-k8s.io/v1alpha1
      kind: YandexMachineTemplate
      name: archtest-control-plane
  replicas: 3
  version: v1.31.4
```



```
apiVersion: infrastructure.cluster.x-k8s.io/v1alpha1
kind: YandexMachineTemplate
metadata:
  name: archtest-control-plane
  namespace: default
spec:
  template:
    spec:
      bootDisk:
        imageID: fd8a3kknu25826s8hbq3 # https://image-builder.sigs.k8s.io/
        size: 100Gi
        typeID: network-ssd
      networkInterfaces:
      - subnetID: e9be3j4n2m1nhp3jneg8
      platformID: standard-v3
      resources:
        cores: 2
        memory: 4Gi
      zoneID: ru-central1-a
```




```
apiVersion: infrastructure.cluster.x-k8s.io/v1alpha1
kind: YandexCluster
metadata:
  name: archtest
  namespace: default
spec:
  folderID: b1g611bc9qj3ajhi18m0
  loadBalancer:
    listener:
      subnet:
        id: e9be3j4n2m1nhp3jneg8
        zoneID: ru-central1-a
      type: ALB
  network:
    id: enp4qpe68i2espgalgdt
```

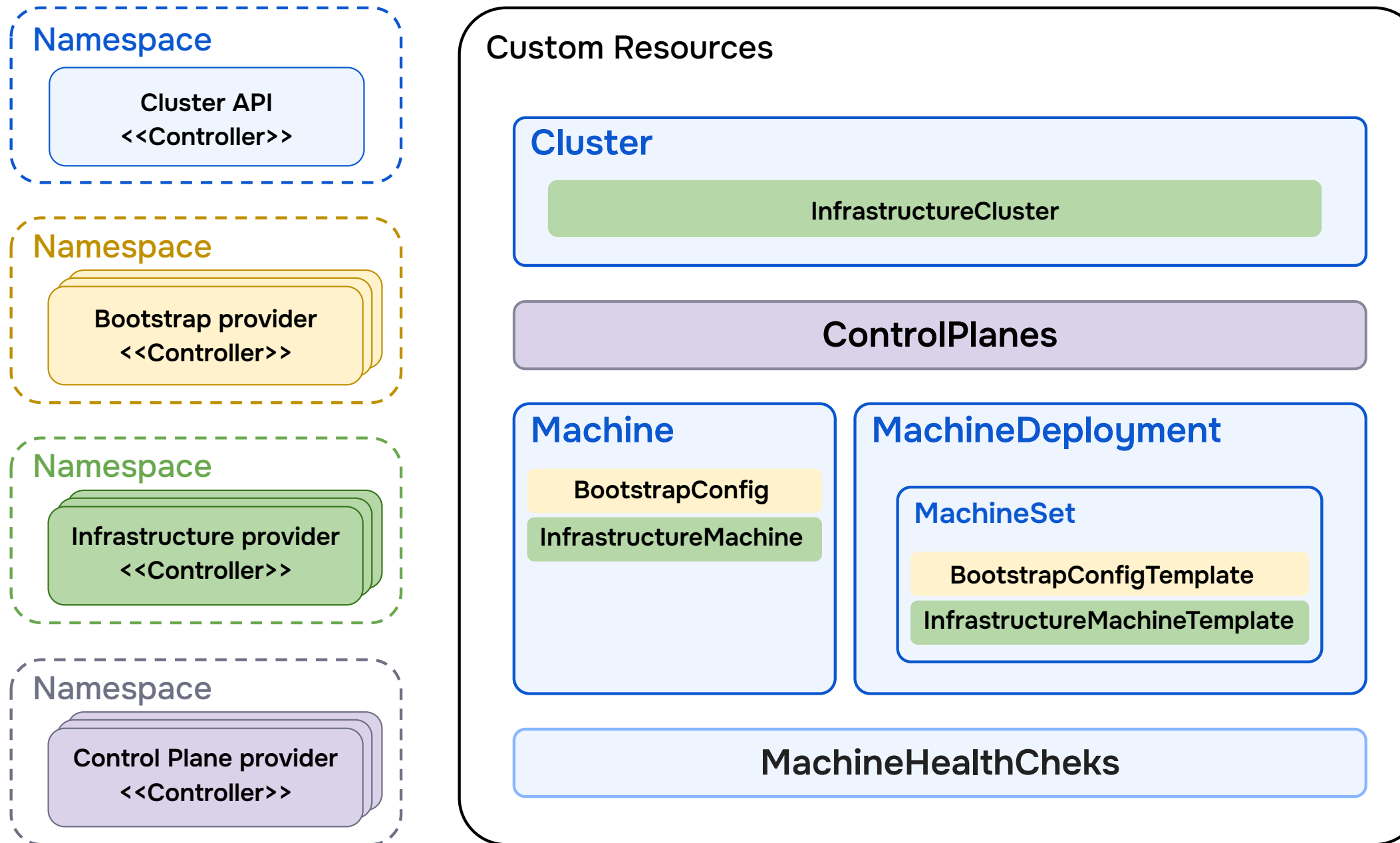


```
apiVersion: cluster.x-k8s.io/v1beta1
kind: MachineDeployment
metadata:
  name: archtest-worker
  namespace: default
spec:
  clusterName: archtest
  replicas: 1
  selector:
    matchLabels:
      role: worker
  template:
    metadata:
      labels:
        cluster.x-k8s.io/cluster-name: archtest
        role: worker
    spec:
      bootstrap:
        configRef:
          apiVersion: bootstrap.cluster.x-k8s.io/v1beta1
          kind: KubeadmConfigTemplate
          name: archtest-worker
      clusterName: archtest
      infrastructureRef:
        apiVersion: infrastructure.cluster.x-k8s.io/v1alpha1
        kind: YandexMachineTemplate
        name: archtest-worker
  version: v1.31.4
```

```
apiVersion: infrastructure.cluster.x-k8s.io/v1alpha1
kind: YandexMachineTemplate
metadata:
  name: archtest-worker
  namespace: default
spec:
  template:
    spec:
      bootDisk:
        imageID: fd8a3kknu25826s8hbq3
        size: 100Gi
        typeID: network-ssd
      networkInterfaces:
        - subnetID: e9be3j4n2m1nhp3jneg8
      platformID: standard-v3
      resources:
        cores: 2
        memory: 4Gi
      zoneID: ru-central1-a
```

```
apiVersion: bootstrap.cluster.x-k8s.io/v1beta1
kind: KubeadmConfigTemplate
metadata:
  name: archtest-worker
  namespace: default
spec:
  template:
    spec:
      joinConfiguration:
        nodeRegistration:
          kubeletExtraArgs:
            cloud-provider: external
            name: '{{ v1.local_hostname }}'
      preKubeadmCommands:
        - hostname '{{ v1.local_hostname }}'
        - echo "::1          ipv6-localhost ipv6-loopback" >/etc/hosts
        - echo "127.0.0.1    localhost" >>/etc/hosts
        - echo "127.0.0.1    {{ v1.local_hostname }}" >>/etc/hosts
        - echo "{{ v1.local_hostname }}" >/etc/hostname
```


Management Cluster Kubernetes



```
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl apply -f archtest-cluster.yaml]
```

Warning: cluster.x-k8s.io/v1beta1 Cluster is deprecated; use cluster.x-k8s.io/v1beta2 Cluster

cluster.cluster.x-k8s.io/archtest created

Warning: controlplane.cluster.x-k8s.io/v1beta1 KubeadmControlPlane is deprecated; use controlplane.cluster.x-k8s.io/v1beta2 KubeadmControlPlane

kubeadmcontrolplane.controlplane.cluster.x-k8s.io/archtest-control-plane created

yandexmachinetemplate.infrastructure.cluster.x-k8s.io/archtest-control-plane created

Warning: cluster.x-k8s.io/v1beta1 MachineDeployment is deprecated; use cluster.x-k8s.io/v1beta2 MachineDeployment

machinedeployment.cluster.x-k8s.io/archtest-worker created

yandexmachinetemplate.infrastructure.cluster.x-k8s.io/archtest-worker created

Warning: bootstrap.cluster.x-k8s.io/v1beta1 KubeadmConfigTemplate is deprecated; use bootstrap.cluster.x-k8s.io/v1beta2 KubeadmConfigTemplate

kubeadmconfigtemplate.bootstrap.cluster.x-k8s.io/archtest-worker created

Error from server (InternalError): error when creating "archtest-cluster.yaml": Internal error occurred: failed calling webhook "mutation.yandexclusters.infrastructure.cluster.x-k8s.io": failed to call webhook: Post "https://capy-webhook-service.capy-system.svc:443/mutate-infrastructure-cluster-x-k8s-io-v1alpha1-yandexcluster-identitylink?timeout=10s": tls: failed to verify certificate: x509: certificate signed by unknown authority

```
kovalkov@archmgmt:~/cluster-api-provider-yandex$
```



Quick Start - The Cluster API Bo X

CNCF Landscape

GitHub - yandex-cloud/cluster-e X

Bug: tls failed to verify certificat X

github.com/yandex-cloud/cluster-api-provider-yandex/issues/45

Sign in

Open

Bug: tls failed to verify certificate: x509: certificate signed by unknown authority #45

kubeadmcontrolplane.controlplane.cluster.x-k8s.io/yandex5-control-plane created
yandexmachinetemplate.infrastructure.cluster.x-k8s.io/yandex5-control-plane created
machinedeployment.cluster.x-k8s.io/yandex5-worker created
yandexmachinetemplate.infrastructure.cluster.x-k8s.io/yandex5-worker created
kubeadmconfigtemplate.bootstrap.cluster.x-k8s.io/yandex5-worker created
Error from server (InternalError): error when creating "/tmp/capy-cluster.yaml": Internal error occurred: failed calling webhook "mutation.yandexclusters.infrastructure.cluster.x-k8s.io": failed to call webhook: Post "https://capy-webhook-service.capy-system.svc:443/mutate-infrastructure-cluster-x-k8s-io-v1alpha1-yandexcluster-identitylink?timeout=10s": tls: failed to verify certificate: x509: certificate signed by unknown authority

sartemov

 added

bug

 on Jul 9, 2025

unreturned

 4 days ago

Quick fix

kubectl annotate MutatingWebhookConfiguration capy-mutating-webhook-configuration cert-manager.io/inject-ca

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

Code with agent mode

No branches or pull requests

Participants

https://github.com/yandex-cloud/cluster-api-provider-yandex/issues/45

kovalkov@archmgmt: ~/cluster-api-provider-yandex — ssh kovalkov@84.252.130.206 — 134x34

```
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl annotate MutatingWebhookConfiguration capy-mutating-webhook-configuration cert-manager.io/inject-ca-from=capy-system/capy-serving-cert
```

```
mutatingwebhookconfiguration.admissionregistration.k8s.io/capy-mutating-webhook-configuration annotated
```

```
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl apply -f archtest-cluster.yaml
```

```
yandexcluster.infrastructure.cluster.x-k8s.io/archtest created
```

```
Warning: cluster.x-k8s.io/v1beta1 Cluster is deprecated; use cluster.x-k8s.io/v1beta2 Cluster
```

```
cluster.cluster.x-k8s.io/archtest configured
```

```
Warning: controlplane.cluster.x-k8s.io/v1beta1 KubeadmControlPlane is deprecated; use controlplane.cluster.x-k8s.io/v1beta2 KubeadmControlPlane
```

```
kubeadmcontrolplane.controlplane.cluster.x-k8s.io/archtest-control-plane unchanged
```

```
yandexmachinetemplate.infrastructure.cluster.x-k8s.io/archtest-control-plane configured
```

```
Warning: cluster.x-k8s.io/v1beta1 MachineDeployment is deprecated; use cluster.x-k8s.io/v1beta2 MachineDeployment
```

```
machinedeployment.cluster.x-k8s.io/archtest-worker configured
```

```
yandexmachinetemplate.infrastructure.cluster.x-k8s.io/archtest-worker configured
```

```
Warning: bootstrap.cluster.x-k8s.io/v1beta1 KubeadmConfigTemplate is deprecated; use bootstrap.cluster.x-k8s.io/v1beta2 KubeadmConfigTemplate
```

```
kubeadmconfigtemplate.bootstrap.cluster.x-k8s.io/archtest-worker unchanged
```

```
kovalkov@archmgmt:~/cluster-api-provider-yandex$
```




```

kovalkov@archmgmt: ~/cluster-api-provider-yandex — ssh kovalkov@84.252.130.206 — 134x34

[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl annotate MutatingWebhookConfiguration capy-mutating-webhook-configuration cert-manager.io/inject-ca-from=capy-system/capy-serving-cert
mutatingwebhookconfiguration.admissionregistration.k8s.io/capy-mutating-webhook-configuration annotated
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl apply -f archtest-cluster.yaml
yandexcluster.infrastructure.cluster.x-k8s.io/archtest created
Warning: cluster.x-k8s.io/v1beta1 Cluster is deprecated; use cluster.x-k8s.io/v1beta2 Cluster
cluster.cluster.x-k8s.io/archtest configured
Warning: controlplane.cluster.x-k8s.io/v1beta1 KubeadmControlPlane is deprecated; use controlplane.cluster.x-k8s.io/v1beta2 KubeadmControlPlane
kubeadmcontrolplane.controlplane.cluster.x-k8s.io/archtest-control-plane unchanged
yandexmachinetemplate.infrastructure.cluster.x-k8s.io/archtest-control-plane configured
Warning: cluster.x-k8s.io/v1beta1 MachineDeployment is deprecated; use cluster.x-k8s.io/v1beta2 MachineDeployment
machinedeployment.cluster.x-k8s.io/archtest-worker configured
yandexmachinetemplate.infrastructure.cluster.x-k8s.io/archtest-worker configured
Warning: bootstrap.cluster.x-k8s.io/v1beta1 KubeadmConfigTemplate is deprecated; use bootstrap.cluster.x-k8s.io/v1beta2 KubeadmConfigTemplate
kubeadmconfigtemplate.bootstrap.cluster.x-k8s.io/archtest-worker unchanged
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl get clusters.cluster.x-k8s.io

```

NAME	CLUSTERCLASS	AVAILABLE	CP DESIRED	CP AVAILABLE	CP UP-TO-DATE	W DESIRED	W AVAILABLE	W UP-TO-DATE	PHASE
archtest		False	3	0	0	1	0	0	Provisionin

```

g 118s
kovalkov@archmgmt:~/cluster-api-provider-yandex$

```




```
kovalkov@archmgmt: ~/cluster-api-provider-yandex — ssh kovalkov@84.252.130.206 — 134x34
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ clusterctl describe cluster archtest
NAME                                REPLICAS AVAILABLE READY UP TO DATE STATUS REASON      SINCE  MESSAGE
Cluster/archtest                    0/4      0      0      0      False  NotAvailable  2m53s  * RemoteC
onnectionProbe: Remote connection not established yet
                                         * Infrast
ructureReady: YandexCluster status.initialization.provisioned is false ←
                                         * Control
PlaneAvailable: Control plane not yet initialized
                                         * Workers
Available:
                                         * Machi
neDeployment archtest-worker: 0 available replicas, at least 1 required
                                         (spec
.strategy.rollout.maxUnavailable is 0, spec.replicas is 1)
└─ClusterInfrastructure - YandexCluster/archtest
└─ControlPlane - KubeadmControlPlane/archtest-control-plane 0/3      0      0      0      False  NotAvailable  2m53s  Control p
lane not yet initialized
kovalkov@archmgmt:~/cluster-api-provider-yandex$
```



```
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl -n capy-system logs --tail 10 capy-controller-manager-7c9df464bd-bfgpn
2026-04-04T23:24:21Z    DEBUG    reconciling application load balancer target group      {"controller": "yandexcluster", "controllerGro
up": "infrastructure.cluster.x-k8s.io", "controllerKind": "YandexCluster", "YandexCluster": {"name": "archtest", "namespace": "default"},
"namespace": "default", "name": "archtest", "reconcileID": "da37cbee-b823-47bd-8a7e-3a54c176a84d"}
2026-04-04T23:24:21Z    DEBUG    reconciling application load balancer backend group      {"controller": "yandexcluster", "controllerGro
up": "infrastructure.cluster.x-k8s.io", "controllerKind": "YandexCluster", "YandexCluster": {"name": "archtest", "namespace": "default"},
"namespace": "default", "name": "archtest", "reconcileID": "da37cbee-b823-47bd-8a7e-3a54c176a84d"}
2026-04-04T23:24:21Z    DEBUG    reconciling application load balancer {"controller": "yandexcluster", "controllerGroup": "infrastruc
ture.cluster.x-k8s.io", "controllerKind": "YandexCluster", "YandexCluster": {"name": "archtest", "namespace": "default"}, "namespace": "d
efault", "name": "archtest", "reconcileID": "da37cbee-b823-47bd-8a7e-3a54c176a84d"}
2026-04-04T23:24:21Z    INFO     load balancer instance not active, requeueing {"controller": "yandexcluster", "controllerGroup": "in
frastructure.cluster.x-k8s.io", "controllerKind": "YandexCluster", "YandexCluster": {"name": "archtest", "namespace": "default"}, "namesp
ace": "default", "name": "archtest", "reconcileID": "da37cbee-b823-47bd-8a7e-3a54c176a84d"}
W0404 23:24:30.208607    1 warnings.go:70] cluster.x-k8s.io/v1beta1 Machine is deprecated; use cluster.x-k8s.io/v1beta2 Machine
2026-04-04T23:24:31Z    INFO     reconciling application load balancer instance {"controller": "yandexcluster", "controllerGroup": "in
frastructure.cluster.x-k8s.io", "controllerKind": "YandexCluster", "YandexCluster": {"name": "archtest", "namespace": "default"}, "namesp
ace": "default", "name": "archtest", "reconcileID": "12a72ed2-ff86-4661-9024-86d16957b096"}
2026-04-04T23:24:31Z    DEBUG    reconciling application load balancer target group      {"controller": "yandexcluster", "controllerGro
up": "infrastructure.cluster.x-k8s.io", "controllerKind": "YandexCluster", "YandexCluster": {"name": "archtest", "namespace": "default"},
"namespace": "default", "name": "archtest", "reconcileID": "12a72ed2-ff86-4661-9024-86d16957b096"}
2026-04-04T23:24:31Z    DEBUG    reconciling application load balancer backend group      {"controller": "yandexcluster", "controllerGro
up": "infrastructure.cluster.x-k8s.io", "controllerKind": "YandexCluster", "YandexCluster": {"name": "archtest", "namespace": "default"},
"namespace": "default", "name": "archtest", "reconcileID": "12a72ed2-ff86-4661-9024-86d16957b096"}
2026-04-04T23:24:31Z    DEBUG    reconciling application load balancer {"controller": "yandexcluster", "controllerGroup": "infrastruc
ture.cluster.x-k8s.io", "controllerKind": "YandexCluster", "YandexCluster": {"name": "archtest", "namespace": "default"}, "namespace": "d
efault", "name": "archtest", "reconcileID": "12a72ed2-ff86-4661-9024-86d16957b096"}
2026-04-04T23:24:31Z    INFO     load balancer instance not active, requeueing {"controller": "yandexcluster", "controllerGroup": "in
frastructure.cluster.x-k8s.io", "controllerKind": "YandexCluster", "YandexCluster": {"name": "archtest", "namespace": "default"}, "namesp
ace": "default", "name": "archtest", "reconcileID": "12a72ed2-ff86-4661-9024-86d16957b096"}
kovalkov@archmgmt:~/cluster-api-provider-yandex$
```




**A FEW
MOMENTS LATER**


```
kovalkov@archmgmt: ~/cluster-api-provider-yandex — ssh kovalkov@84.252.130.206 — 134x34

[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl get clusters
NAME          CLUSTERCLASS  AVAILABLE  CP DESIRED  CP AVAILABLE  CP UP-TO-DATE  W DESIRED  W AVAILABLE  W UP-TO-DATE  PHASE
AGE  VERSION
archtest      False        3          0           1             1             0          1            Provisioned
10m

[kovalkov@archmgmt:~/cluster-api-provider-yandex$ clusterctl describe cluster archtest
NAME          REPLICAS AVAILABLE  READY UP TO DATE  STATUS  REASON
SINCE  MESSAGE
Cluster/archtest      2/4      0      0      2      False  NotAvailable
11m    * ControlPlaneAvailable: Failed to get etcd members

      * WorkersAvailable:

        * MachineDeployment archtest-worker: 0 available replicas, at least 1 required

          (spec.strategy.rollout.maxUnavailable is 0, spec.replicas is 1)
ClusterInfrastructure - YandexCluster/archtest

ControlPlane - KubeadmControlPlane/archtest-control-plane      1/3      0      0      1      Unknown  InspectionFail
ed 14s    Failed to get etcd members
Machine/archtest-control-plane-rwclh      1      0      0      1      Unknown  ReadyUnknown
3m13s    * NodeHealthy: Waiting for a Node with spec.providerID yandex://fhmlh1r2ucpcb017rep4 to exist

      * Control plane components: Waiting for a Node with spec.providerID yandex://fhmlh1r2ucpcb017rep4 to
        exist

      * EtcdMemberHealthy: Waiting for a Node with spec.providerID yandex://fhmlh1r2ucpcb017rep4 to exist
MachineInfrastructure - YandexMachine/archtest-control-plane-rwclh

Workers

MachineDeployment/archtest-worker      1/1      0      0      1      False  NotAvailable
11m    0 available replicas, at least 1 required (spec.strategy.rollout.maxUnavailable is 0, spec.replicas
```




```
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ clusterctl get kubeconfig archtest > /home/kovalkov/.kube/archtest
```

```
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl --kubeconfig /home/kovalkov/.kube/archtest get pods --all-namespaces
```

NAMESPACE	NAME	READY	STATUS	RESTARTS	AGE
kube-system	coredns-7c65d6cfc9-bp8mt	0/1	Pending	0	4m42s
kube-system	coredns-7c65d6cfc9-w6jss	0/1	Pending	0	4m42s
kube-system	etcd-archtest-control-plane-rwclh	1/1	Running	0	5m7s
kube-system	kube-apiserver-archtest-control-plane-rwclh	1/1	Running	0	5m3s
kube-system	kube-controller-manager-archtest-control-plane-rwclh	1/1	Running	0	4m53s
kube-system	kube-proxy-4spzw	1/1	Running	0	2m57s
kube-system	kube-proxy-97t64	1/1	Running	0	4m43s
kube-system	kube-scheduler-archtest-control-plane-rwclh	1/1	Running	0	4m53s

```
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl --kubeconfig /home/kovalkov/.kube/archtest get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
archtest-control-plane-rwclh	NotReady	control-plane	5m15s	v1.31.4
archtest-worker-zk8kf-qn2bn	NotReady	<none>	3m3s	v1.31.4

```
kovalkov@archmgmt:~/cluster-api-provider-yandex$
```



```
kovalkov — kovalkov@archmgmt: ~ — ssh kovalkov@84.252.130.206 — 134x34

[kovalkov@archmgmt:~$ kubectl --kubeconfig /home/kovalkov/.kube/archtest --namespace kube-system describe pods coredns-7c65d6cfc9-2jdss]
| tail -n 4
Events:
  Type            Reason              Age             From              Message
  ----            -
Warning FailedScheduling  3m35s (x3 over 14m)  default-scheduler  0/1 nodes are available: 1 node(s) had intolerated taint {node.cloudprovider.kubernetes.io/uninitialized: true}. Preemption is not helpful for scheduling.

[kovalkov@archmgmt:~$ kubectl --kubeconfig /home/kovalkov/.kube/archtest describe nodes archtest-control-plane-jq8md | grep '^Conditions' -A 6]
Conditions:
  Type            Status  LastHeartbeatTime               LastTransitionTime              Reason                               Message
  ----            -
MemoryPressure   False   Mon, 06 Apr 2026 22:33:25 +0000   Mon, 06 Apr 2026 22:22:55 +0000   KubeletHasSufficientMemory          kubelet has sufficient memory available
DiskPressure     False   Mon, 06 Apr 2026 22:33:25 +0000   Mon, 06 Apr 2026 22:22:55 +0000   KubeletHasNoDiskPressure            kubelet has no disk pressure
PIDPressure      False   Mon, 06 Apr 2026 22:33:25 +0000   Mon, 06 Apr 2026 22:22:55 +0000   KubeletHasSufficientPID             kubelet has sufficient PID available
Ready            False   Mon, 06 Apr 2026 22:33:25 +0000   Mon, 06 Apr 2026 22:22:55 +0000   KubeletNotReady                     container runtime network not ready: NetworkReady=false reason:NetworkPluginNotReady message:Network plugin returns error: cni plugin not initialized

kovalkov@archmgmt:~$
```


Quick Start - The Cluster API Bo X CNCF Landscape X GitHub - yandex-cloud/cluster-api-provider-yandex X Bug: tls failed to verify certificat X + Private browsing

github.com/yandex-cloud/cluster-api-provider-yandex Sign in

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Установите в созданный кластер CCSM

Для обеспечения связи между ресурсами кластера и ресурсами Yandex Cloud, установите в созданный кластер [Cloud Controller Manager](#), например [Kubernetes Cloud Controller Manager for Yandex Cloud](#).

Important

Если вы хотите использовать решение Kubernetes Cloud Controller Manager for Yandex Cloud, добавьте в пример манифеста [yandex-cloud-controller-manager.yaml](#) для `DaemonSet` актуальную версию Docker-образа и переменную окружения `YANDEX_CLUSTER_NAME` с именем созданного кластера в значении.

Установите в созданный кластер CNI

Чтобы обеспечить сетевую функциональность для подов в новом кластере, установите в него [Container Network Interface](#), например [Cilium](#) или [Calico](#).

Подробнее см. в документации:

- [Cilium Quick Installation](#)
- [Quickstart for Calico on Kubernetes](#)

Проверьте связь управляющего кластера с созданным

<https://github.com/yandex-cloud/cluster-api-provider-yandex/>

Quick Start - The Cluster API Bo X

CNCF Landscape

GitHub - yandex-cloud/cluster-e X

Cilium Quick Installation — Ciliu X

docs.cilium.io/en/stable/gettingstarted/k8s-install-default/#install-the-cilium-cli

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Running Prometheus & Grafana

Monitoring & Metrics

Layer 7 Protocol Visibility

OPERATIONS

System Requirements

Upgrade Guide

Configuration

Performance & Scalability

Troubleshooting

COMMUNITY

Governance

LinuxmacOSOther

```
CILIUM_CLI_VERSION=$(curl -s https://raw.githubusercontent.com/cilium/cilium-cli/main/stable.  
CLI_ARCH=amd64  
if [ "$(uname -m)" = "aarch64" ]; then CLI_ARCH=arm64; fi  
curl -L --fail --remote-name-all https://github.com/cilium/cilium-cli/releases/download/${CIL  
sha256sum --check cilium-linux-${CLI_ARCH}.tar.gz.sha256sum  
sudo tar xzvfC cilium-linux-${CLI_ARCH}.tar.gz /usr/local/bin  
rm cilium-linux-${CLI_ARCH}.tar.gz{,.sha256sum}
```

Video

To learn more about the Cilium CLI, check out [eCHO episode 8: Exploring the Cilium CLI](#).

Read the Docsversion: stable

<https://docs.cilium.io/en/stable/gettingstarted/k8s-install-default/#install-the-cilium-cli>

Quick Start - The Cluster API Bo X

CNCF Landscape

GitHub - yandex-cloud/cluster-e X

GitHub - deckhouse/yandex-clo X

+

▼

Private browsing

github.com/deckhouse/yandex-cloud-controller-manager

Sign in

go.sum

Fix CVE (#193)

last month

README

Apache-2.0 license

Kubernetes Cloud Controller Manager for Yandex.Cloud

CI failing

go report A+

Overview

yandex-cloud-controller-manager is the Kubernetes Cloud Controller Manager (CCM) implementation for Yandex.Cloud. It allows you to leverage many of the cloud provider features offered by Yandex.Cloud on your Kubernetes clusters. Read more about Kubernetes CCM [here](#).

Currently yandex-cloud-controller-manager implements:

- NodeController - responsible for updating kubernetes nodes with cloud provider specific labels and addresses and deleting kubernetes nodes that were deleted on your cloud.
- ServiceController - responsible for creating LoadBalancers when a service of Type: LoadBalancer is created in Kubernetes.
- RouteController - responsible for creating routes to allow networking between Pods.

Contributors 17



[+ 3 contributors](#)

Languages

Go 91.9%

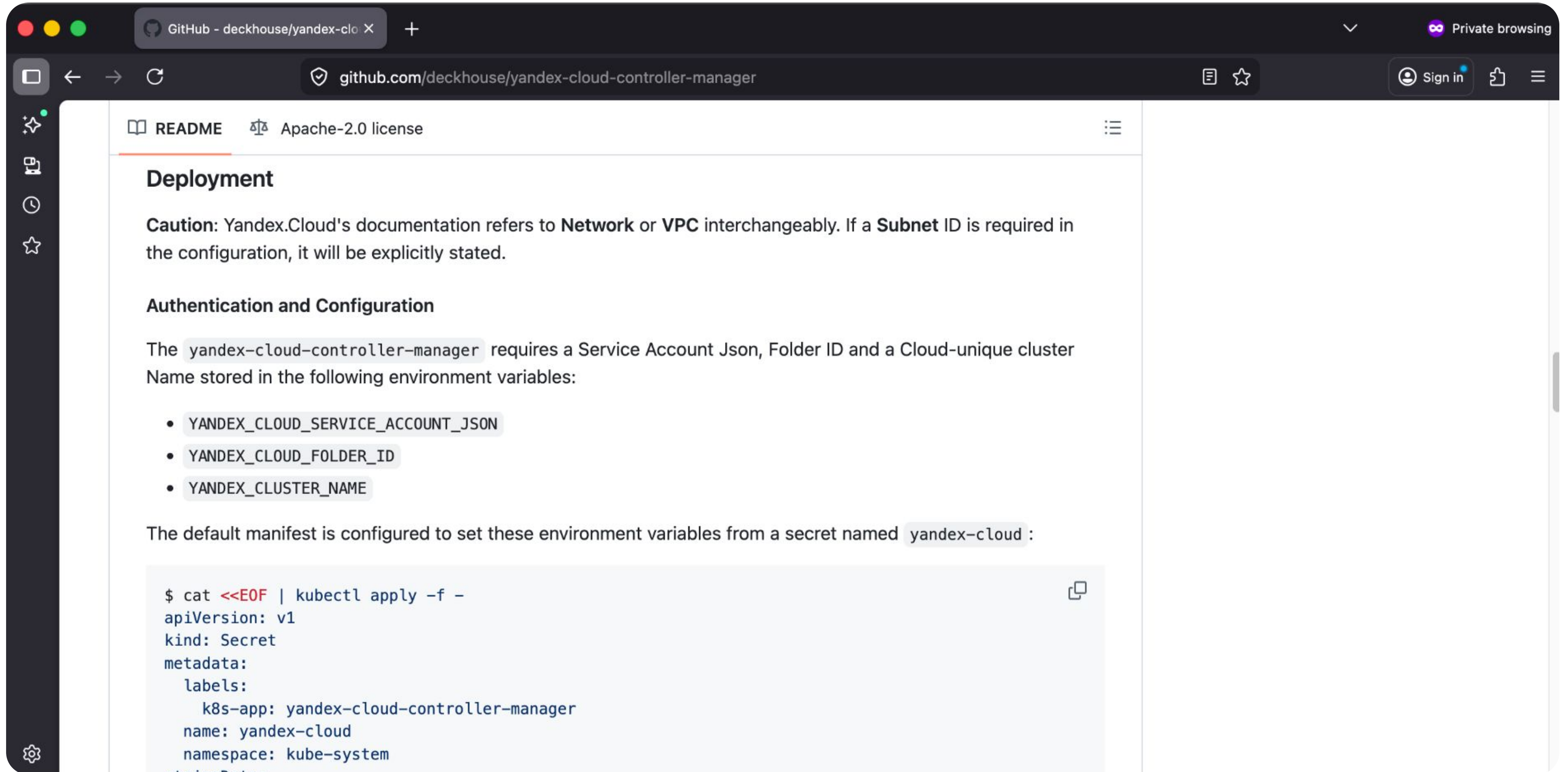
Go Template 2.9%

Makefile 2.0%

Dockerfile 1.8%

Shell 1.4%

<https://github.com/deckhouse/yandex-cloud-controller-manager>




```
kovalkov@archmgmt: ~/yandex-cloud-controller-manager — ssh kovalkov@84.252.130.206 — 134x34

[kovalkov@archmgmt:~$ cilium --kubeconfig /home/kovalkov/.kube/archtest install
i Using Cilium version 1.19.1
🔮 Auto-detected cluster name: archtest
🔮 Auto-detected kube-proxy has been installed
[kovalkov@archmgmt:~$ git clone https://github.com/deckhouse/yandex-cloud-controller-manager.git
Cloning into 'yandex-cloud-controller-manager'...
remote: Enumerating objects: 4933, done.
remote: Counting objects: 100% (191/191), done.
remote: Compressing objects: 100% (127/127), done.
remote: Total 4933 (delta 139), reused 71 (delta 57), pack-reused 4742 (from 4)
Receiving objects: 100% (4933/4933), 7.80 MiB | 14.61 MiB/s, done.
Resolving deltas: 100% (1699/1699), done.
[kovalkov@archmgmt:~$ cd yandex-cloud-controller-manager/
[kovalkov@archmgmt:~/yandex-cloud-controller-manager$ vim yandex-cloud.yaml
[kovalkov@archmgmt:~/yandex-cloud-controller-manager$ kubectl --kubeconfig /home/kovalkov/.kube/archtest create -f yandex-cloud.yaml
secret/yandex-cloud created
[kovalkov@archmgmt:~/yandex-cloud-controller-manager$ kubectl --kubeconfig /home/kovalkov/.kube/archtest apply -f manifests/yandex-cloud-controller-manager-rbac.yaml
clusterrole.rbac.authorization.k8s.io/system:cloud-controller-manager created
clusterrolebinding.rbac.authorization.k8s.io/system:cloud-controller-manager created
[kovalkov@archmgmt:~/yandex-cloud-controller-manager$ kubectl --kubeconfig /home/kovalkov/.kube/archtest apply -f manifests/yandex-cloud-controller-manager.yaml
serviceaccount/cloud-controller-manager created
daemonset.apps/yandex-cloud-controller-manager created
[kovalkov@archmgmt:~/yandex-cloud-controller-manager$ kubectl --kubeconfig /home/kovalkov/.kube/archtest -n kube-system edit daemonsets/.apps yandex-cloud-controller-manager
daemonset.apps/yandex-cloud-controller-manager edited
kovalkov@archmgmt:~/yandex-cloud-controller-manager$
```



```
creationTimestamp: "2026-04-04T23:43:15Z"
generation: 1
labels:
  k8s-app: yandex-cloud-controller-manager
name: yandex-cloud-controller-manager
namespace: kube-system
resourceVersion: "2052"
uid: 394d2799-9baa-41b9-9283-44c290f165e7
spec:
  revisionHistoryLimit: 10
  selector:
    matchLabels:
      k8s-app: yandex-cloud-controller-manager
  template:
    metadata:
      creationTimestamp: null
      labels:
        k8s-app: yandex-cloud-controller-manager
    spec:
      containers:
      - command:
        - /bin/yandex-cloud-controller-manager
        - --cloud-provider=yandex
        - --v=3
        env:
        - name: YANDEX_CLUSTER_NAME
          value: archtest
        - name: YANDEX_CLOUD_SERVICE_ACCOUNT_JSON
          valueFrom:
            secretKeyRef:
              key: service-account-json
              name: yandex-cloud
        - name: YANDEX_CLOUD_FOLDER_ID
```



[kovalkov@archmgmt:~\$ kubectl --kubeconfig /home/kovalkov/.kube/archtest get pods --all-namespaces

NAMESPACE	NAME	READY	STATUS	RESTARTS	AGE
kube-system	cilium-445wv	1/1	Running	0	9m43s
kube-system	cilium-c5xqw	1/1	Running	0	9m43s
kube-system	cilium-envoy-g7lgb	1/1	Running	0	84s
kube-system	cilium-envoy-j4v4s	1/1	Running	0	9m43s
kube-system	cilium-envoy-ll2jl	1/1	Running	0	9m43s
kube-system	cilium-lch97	1/1	Running	0	84s
kube-system	cilium-operator-6f7d5bfd95-h8hbt	1/1	Running	0	9m43s
kube-system	coredns-7c65d6cfc9-bp8mt	1/1	Running	0	18m
kube-system	coredns-7c65d6cfc9-w6jss	1/1	Running	0	18m
kube-system	etcd-archtest-control-plane-pm4vz	1/1	Running	0	21s
kube-system	etcd-archtest-control-plane-rwclh	1/1	Running	0	18m
kube-system	kube-apiserver-archtest-control-plane-pm4vz	1/1	Running	0	10s
kube-system	kube-apiserver-archtest-control-plane-rwclh	1/1	Running	0	18m
kube-system	kube-controller-manager-archtest-control-plane-pm4vz	1/1	Running	0	18s
kube-system	kube-controller-manager-archtest-control-plane-rwclh	1/1	Running	0	18m
kube-system	kube-proxy-4spzw	1/1	Running	0	16m
kube-system	kube-proxy-97t64	1/1	Running	0	18m
kube-system	kube-proxy-bgwx2	1/1	Running	0	84s
kube-system	kube-scheduler-archtest-control-plane-pm4vz	1/1	Running	0	82s
kube-system	kube-scheduler-archtest-control-plane-rwclh	1/1	Running	0	18m
kube-system	yandex-cloud-controller-manager-4ntnk	1/1	Running	0	2m52s
kube-system	yandex-cloud-controller-manager-q2hq5	1/1	Running	0	20s

[kovalkov@archmgmt:~\$ kubectl --kubeconfig /home/kovalkov/.kube/archtest get nodes

NAME	STATUS	ROLES	AGE	VERSION
archtest-control-plane-pm4vz	Ready	control-plane	86s	v1.31.4
archtest-control-plane-rwclh	Ready	control-plane	18m	v1.31.4
archtest-worker-zk8kf-qn2bn	Ready	<none>	16m	v1.31.4



kovalkov@archmgmt:~\$


```
kovalkov@archmgmt: ~ — ssh kovalkov@84.252.130.206 — 134x34
[kovalkov@archmgmt:~/yandex-cloud-controller-manager$ cd
[kovalkov@archmgmt:~$ kubectl get clusters
NAME          CLUSTERCLASS  AVAILABLE  CP DESIRED  CP AVAILABLE  CP UP-TO-DATE  W DESIRED  W AVAILABLE  W UP-TO-DATE  PHASE
AGE  VERSION
archtest      True         3          1           2             1             1          1            1             ed
25m
[kovalkov@archmgmt:~$ clusterctl describe cluster archtest
NAME          REPLICAS AVAILABLE READY UP TO DATE STATUS REASON  SINCE
MESSAGE
Cluster/archtest 3/4      2          2          3          True  Available 27s
* ControlPlaneAvailable:
  * 1 of 2 Machines has healthy control plane components, at least 1 required
ClusterInfrastructure - YandexCluster/archtest
ControlPlane - KubeadmControlPlane/archtest-control-plane 2/3      1          1          2          True  Available 27s
  * 1 of 2 Machines has healthy control plane components, at least 1 required
Machine/archtest-control-plane-pm4vz 1         0          0          1          False NotReady 27s
* NodeHealthy:
  * Node.Ready: container runtime network not ready: NetworkReady=false reason:NetworkPluginNotReady
    message:Network plugin returns error: cni plugin not initialized
  * APIServerPodHealthy: Pod does not exist
  * ControllerManagerPodHealthy: Pod does not exist
  * SchedulerPodHealthy: Waiting for startup or readiness probes
  * EtcdPodHealthy: Pod does not exist
MachineInfrastructure - YandexMachine/archtest-control-plane-pm4vz
Machine/archtest-control-plane-rwclh 1         1          1          1          True  Ready 112s
```




**A FEW
MOMENTS LATER**


```
[kovalkov@archmgmt:~$ kubectl get clusters
```

NAME	CLUSTERCLASS	AVAILABLE	CP DESIRED	CP AVAILABLE	CP UP-TO-DATE	W DESIRED	W AVAILABLE	W UP-TO-DATE	PHASE
AGE	VERSION								
archtest		True	3	3	3	1	1	1	Provisioned
37m									

```
[kovalkov@archmgmt:~$ clusterctl describe cluster archtest
```

NAME	REPLICAS	AVAILABLE	READY	UP TO DATE	STATUS	REASON	SINCE
MESSAGE							
Cluster/ archtest	4/4	4	4	4	True	Available	12m

```
└─ClusterInfrastructure - YandexCluster/archtest
```

```
└─ControlPlane - KubeadmControlPlane/archtest-control-plane 3/3 3 3 3 True Available 12m
```

```
└─ 3 Machines... 3 3 3 True Ready 13m
```

```
See archtest-control-plane-2px9r, archtest-control-plane-pm4vz, ...
```

```
└─Workers
```

```
└─MachineDeployment/archtest-worker 1/1 1 1 1 True Available 13m
```

```
└─Machine/archtest-worker-zk8kf-qn2bn 1 1 1 1 True Ready 13m
```

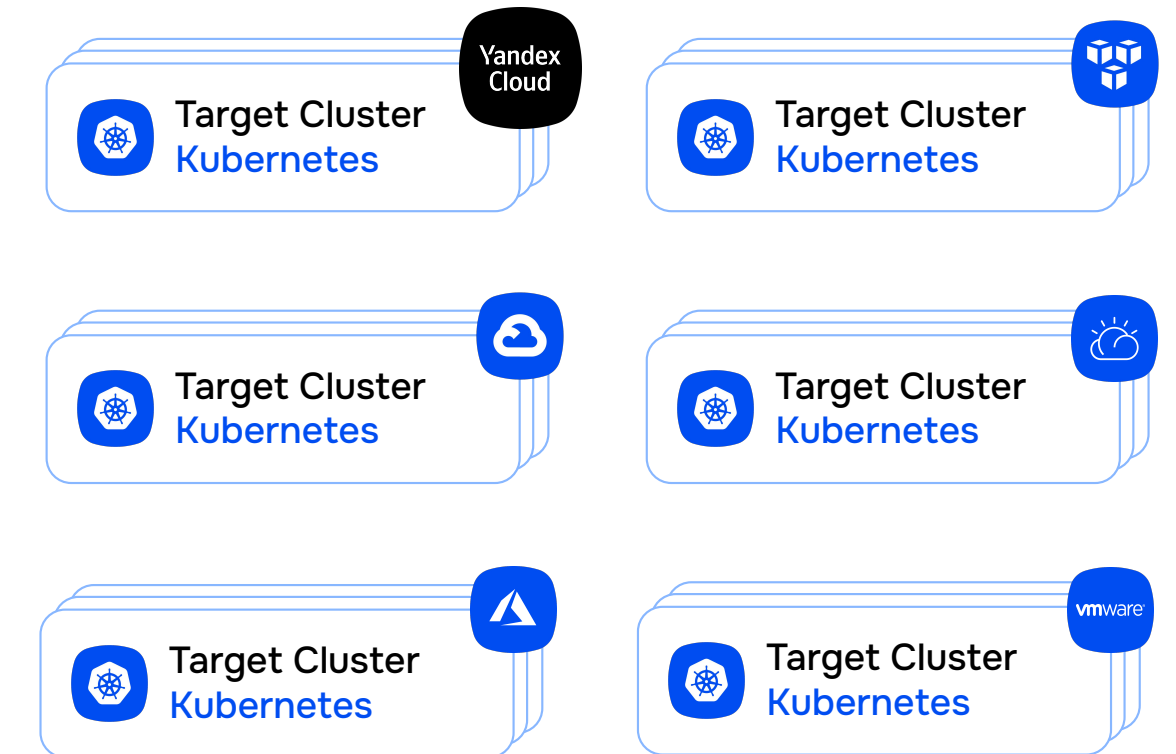
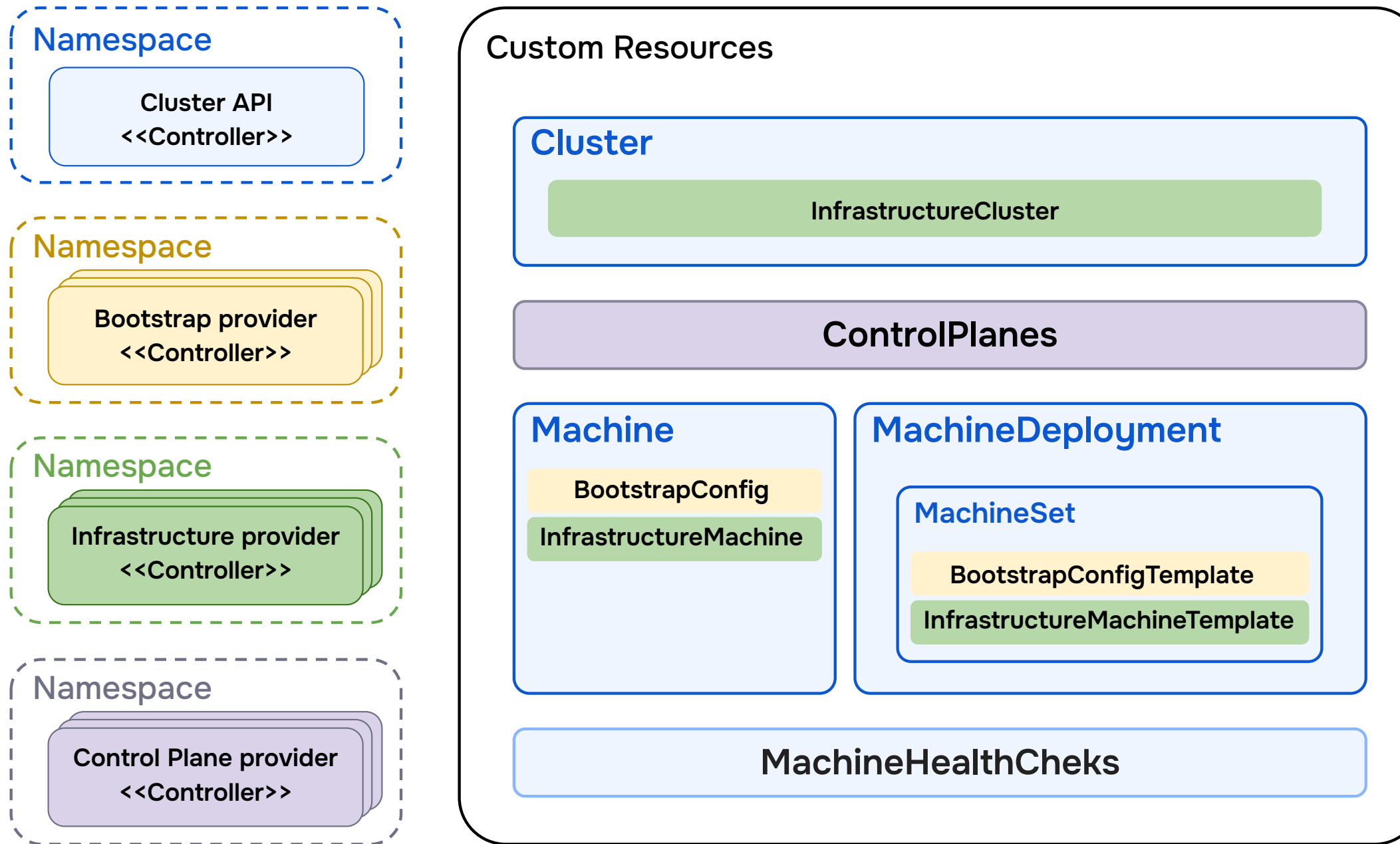
```
└─MachineInfrastructure - YandexMachine/archtest-worker-zk8kf-qn2bn
```

```
[kovalkov@archmgmt:~$ kubectl --kubeconfig /home/kovalkov/.kube/archtest get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
archtest-control-plane-2px9r	Ready	control-plane	8m8s	v1.31.4
archtest-control-plane-pm4vz	Ready	control-plane	13m	v1.31.4
archtest-control-plane-rwclh	Ready	control-plane	30m	v1.31.4
archtest-worker-zk8kf-qn2bn	Ready	<none>	28m	v1.31.4

```
kovalkov@archmgmt:~$
```


Management Cluster Kubernetes




```
    cloud-provider: external
    name: '{{ v1.local_hostname }}'
joinConfiguration:
  controlPlane:
    localAPIEndpoint:
      bindPort: 8443
  nodeRegistration:
    kubeletExtraArgs:
      cloud-provider: external
      name: '{{ v1.local_hostname }}'
preKubeadmCommands:
- hostname "{{ v1.local_hostname }}"
- echo "::1          ipv6-localhost ipv6-loopback" >/etc/hosts
- echo "127.0.0.1    localhost" >>/etc/hosts
- echo "127.0.0.1    {{ v1.local_hostname }}" >>/etc/hosts
- echo "{{ v1.local_hostname }}" >/etc/hostname
machineTemplate:
  infrastructureRef:
    apiVersion: infrastructure.cluster.x-k8s.io/v1alpha1
    kind: YandexMachineTemplate
    name: archtest-control-plane
    replicas: 1
    version: v1.31.4
  --
apiVersion: infrastructure.cluster.x-k8s.io/v1alpha1
kind: YandexMachineTemplate
metadata:
  name: archtest-control-plane
  namespace: default
spec:
  template:
    spec:
      bootDisk:
-- INSERT --
```



```
kovalkov@archmgmt: ~/cluster-api-provider-yandex — ssh kovalkov@84.252.130.206 — 134x34
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ vim archtest-cluster.yaml
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl apply -f archtest-cluster.yaml
yandexcluster.infrastructure.cluster.x-k8s.io/archtest unchanged
Warning: cluster.x-k8s.io/v1beta1 Cluster is deprecated; use cluster.x-k8s.io/v1beta2 Cluster
cluster.cluster.x-k8s.io/archtest configured
Warning: controlplane.cluster.x-k8s.io/v1beta1 KubeadmControlPlane is deprecated; use controlplane.cluster.x-k8s.io/v1beta2 KubeadmControlPlane
kubeadmcontrolplane.controlplane.cluster.x-k8s.io/archtest-control-plane configured
yandexmachinetemplate.infrastructure.cluster.x-k8s.io/archtest-control-plane configured
Warning: cluster.x-k8s.io/v1beta1 MachineDeployment is deprecated; use cluster.x-k8s.io/v1beta2 MachineDeployment
machinedeployment.cluster.x-k8s.io/archtest-worker configured
yandexmachinetemplate.infrastructure.cluster.x-k8s.io/archtest-worker configured
Warning: bootstrap.cluster.x-k8s.io/v1beta1 KubeadmConfigTemplate is deprecated; use bootstrap.cluster.x-k8s.io/v1beta2 KubeadmConfigTemplate
kubeadmconfigtemplate.bootstrap.cluster.x-k8s.io/archtest-worker unchanged
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl get clusters
```

NAME	CLUSTERCLASS	AVAILABLE	CP DESIRED	CP AVAILABLE	CP UP-TO-DATE	W DESIRED	W AVAILABLE	W UP-TO-DATE	PHASE
archtest		True	1	2	3	1	1	1	Provisioned

```
kovalkov@archmgmt:~/cluster-api-provider-yandex$
```




```
kovalkov@archmgmt: ~/cluster-api-provider-yandex — ssh kovalkov@84.252.130.206 — 134x34
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ clusterctl describe cluster archtest
NAME
SON    SINCE MESSAGE
Cluster/archtest
ilable 15m * ControlPlaneAvailable:
|
|         * 2 of 3 etcd members are healthy, at least 2 healthy member required for etcd quorum
|
|         * 2 of 3 Machines have healthy control plane components, at least 1 required
|ClusterInfrastructure - YandexCluster/archtest
|ControlPlane - KubeadmControlPlane/archtest-control-plane
ilable 15m * 2 of 3 etcd members are healthy, at least 2 healthy member required for etcd quorum
|
|         * 2 of 3 Machines have healthy control plane components, at least 1 required
|!! DELETED !! Machine/archtest-control-plane-rwclh
Ready 36s * Deleting: Machine deletion in progress, stage: WaitingForInfrastructureDeletion
|
|         * Control plane components: Machine is deleting
|
|         * EtcdMemberHealthy: Etcd doesn't have an etcd member for Node archtest-control-plane-rwclh
|!! DELETED !! MachineInfrastructure - YandexMachine/archtest-control-plane-rwclh
|2 Machines...
dy 14m See archtest-control-plane-2px9r, archtest-control-plane-pm4vz
Workers
MachineDeployment/archtest-worker
ilable 16m
Machine/archtest-worker-zk8kf-qn2bn
dy 16m
MachineInfrastructure - YandexMachine/archtest-worker-zk8kf-qn2bn
kovalkov@archmgmt:~/cluster-api-provider-yandex$
```





**A FEW
MOMENTS LATER**


```
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl get clusters
```

NAME	CLUSTERCLASS	AVAILABLE	CP DESIRED	CP AVAILABLE	CP UP-TO-DATE	W DESIRED	W AVAILABLE	W UP-TO-DATE	PHASE
archtest		True	1	1	1	1	1	1	Provisioned

AGE VERSION

43m

```
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ clusterctl describe cluster archtest
```

NAME	REPLICAS	AVAILABLE	READY	UP TO DATE	STATUS	REASON	SINCE
Cluster/archtest	2/2	2	2	2	True	Available	2m16s

```
└─ClusterInfrastructure - YandexCluster/archtest
```

```
└─ControlPlane - KubeadmControlPlane/archtest-control-plane
```

REPLICAS	AVAILABLE	READY	UP TO DATE	STATUS	REASON	SINCE
1/1	1	1	1	True	Available	2m17s

```
└─┬─Machine/archtest-control-plane-2px9r
```

REPLICAS	AVAILABLE	READY	UP TO DATE	STATUS	REASON	SINCE
1	1	1	1	True	Ready	2m17s

```
└─┬─┬─MachineInfrastructure - YandexMachine/archtest-control-plane-2px9r
```

```
└─┬─Workers
```

```
└─┬─┬─MachineDeployment/archtest-worker
```

REPLICAS	AVAILABLE	READY	UP TO DATE	STATUS	REASON	SINCE
1/1	1	1	1	True	Available	19m

```
└─┬─┬─┬─Machine/archtest-worker-zk8kf-qn2bn
```

REPLICAS	AVAILABLE	READY	UP TO DATE	STATUS	REASON	SINCE
1	1	1	1	True	Ready	19m

```
└─┬─┬─┬─┬─MachineInfrastructure - YandexMachine/archtest-worker-zk8kf-qn2bn
```

```
[kovalkov@archmgmt:~/cluster-api-provider-yandex$ kubectl --kubeconfig /home/kovalkov/.kube/archtest get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
archtest-control-plane-2px9r	Ready	control-plane	13m	v1.31.4
archtest-worker-zk8kf-qn2bn	Ready	<none>	33m	v1.31.4

```
kovalkov@archmgmt:~/cluster-api-provider-yandex$
```


[kovalkov@archmgmt:~/cluster-api-provider-yandex\$ clusterctl describe cluster archtest

NAME	REPLICAS	AVAILABLE	READY	UP TO DATE	STATUS	REASON	SINCE
Cluster/ archtest	1/1	1	1	1	True	Available	8m31s
└─ClusterInfrastructure - YandexCluster/archtest							
└─ControlPlane - KubeadmControlPlane/archtest-control-plane	1/1	1	1	1	True	Available	8m32s
└─Machine/archtest-control-plane-2px9r	1	1	1	1	True	Ready	8m32s
└─MachineInfrastructure - YandexMachine/archtest-control-plane-2px9r							

[kovalkov@archmgmt:~/cluster-api-provider-yandex\$ kubectl --kubeconfig /home/kovalkov/.kube/archtest get nodes

NAME	STATUS	ROLES	AGE	VERSION
archtest-control-plane-2px9r	Ready	control-plane	19m	v1.31.4

[kovalkov@archmgmt:~/cluster-api-provider-yandex\$ kubectl --kubeconfig /home/kovalkov/.kube/archtest get pods --all-namespaces

NAMESPACE	NAME	READY	STATUS	RESTARTS	AGE
kube-system	cilium-envoy-w795l	1/1	Running	0	20m
kube-system	cilium-operator-6f7d5bfd95-srmrr	1/1	Running	0	4m35s
kube-system	cilium-p8nn2	1/1	Running	0	20m
kube-system	coredns-7c65d6cfc9-rt8gp	1/1	Running	0	4m35s
kube-system	coredns-7c65d6cfc9-t7c95	1/1	Running	0	10m
kube-system	etcd-archtest-control-plane-2px9r	1/1	Running	0	18m
kube-system	kube-apiserver-archtest-control-plane-2px9r	1/1	Running	1 (20m ago)	18m
kube-system	kube-controller-manager-archtest-control-plane-2px9r	1/1	Running	0	18m
kube-system	kube-proxy-t5gr4	1/1	Running	0	20m
kube-system	kube-scheduler-archtest-control-plane-2px9r	1/1	Running	0	18m
kube-system	yandex-cloud-controller-manager-t9vg4	1/1	Running	0	18m

kovalkov@archmgmt:~/cluster-api-provider-yandex\$

CNI: Cilium/Calico/etc?

Security: Kyverno/Gatekeeper/etc?

Auth: Dex/Keycloak/etc?

Storage?

Auto-scaling?

Monitoring: Prometheus/VictoriaMetrics/etc?

Logs: Fluentd/Vector/etc?



ИТОГИ

- ✓ Management Cluster +1

<https://github.com/k0sproject/k0s>

- ✓ +2 Contribution on GitHub

<https://github.com/yandex-cloud/cluster-api-provider-yandex/issues/48>

<https://github.com/yandex-cloud/cluster-api-provider-yandex/issues/45>

- ✓ Время PoC: ~3 вечера

- ✓ Специальный образ ОС
Warnings, Errors, CVEs





Deckhouse
ФЛАНТ

Quick Start - The Cluster AI X

CNCF Landscape X

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+

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Deckhouse Kubernetes Platform

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discussions

14 total

Contributor Covenant

2.1

releases

releases.deckhouse.io

[Deckhouse Kubernetes Platform](#) is an Open Source platform for managing Kubernetes clusters in a fully automatic and uniform fashion. It allows you to create homogeneous Kubernetes clusters anywhere and fully manages them. It supplies all the add-ons you need for auto-scaling, observability, security, and service mesh. It comes in Enterprise Edition (EE) and Community Edition (CE).

Main features

- NoOps: system software on the nodes, Kubernetes core software, Kubernetes platform components are automatically managed.
- SLA by design: availability can be guaranteed even without direct access to your infrastructure.

certified

<https://github.com/deckhouse/deckhouse>

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RU

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1

2

3

4

5

6

Выбор
инфраструктуры

Информация об
установке

Подготовка
окружения

Установка

Получаем доступ
к кластеру

Куда двигаться
дальше?

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ДОКУМЕНТАЦИЯ СВЯЖИТЕСЬ С НАМИ RU

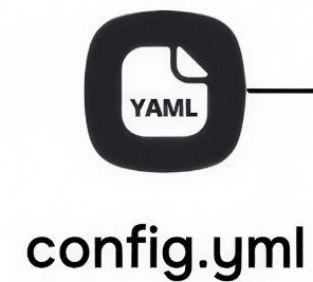
Kubernetes Platform О продукте Возможности ФСТЭК-редакция Лицензирование Сценарии использования Быстрый старт Документация Модули Руководства

Создайте файл `config.yml`.

```
1 # Общие параметры кластера.
2 # https://deckhouse.ru/products/kubernetes-platform/documentation/v1/reference/api/cr.html#clusterconfiguration
3 apiVersion: deckhouse.io/v1
4 kind: ClusterConfiguration
5 clusterType: Cloud
6 cloud:
7   provider: Yandex
8   # Префикс объектов, создаваемых в облаке при установке.
9   # Возможно, захотите изменить.
10  prefix: cloud-demo
11 # Адресное пространство Pod'ов кластера.
12 # Возможно, захотите изменить. Убедитесь, что не будет пересечений с serviceSubnetCIDR.
13 podSubnetCIDR: 10.111.0.0/16
14 # Адресное пространство для service'ов кластера.
15 # Возможно, захотите изменить. Убедитесь, что не будет пересечений с podSubnetCIDR.
16 serviceSubnetCIDR: 10.222.0.0/16
17 kubernetesVersion: "Automatic"
18 clusterDomain: "cluster.local"
19 defaultCPI: "ContainerdV2"
```

<https://deckhouse.ru/products/kubernetes-platform/gs/yandex/step4.html>


```
apiVersion: deckhouse.io/v1
kind: ClusterConfiguration
clusterType: Cloud
cloud:
  provider: Yandex
  prefix: cloud-demo
podSubnetCIDR: 10.111.0.0/16
serviceSubnetCIDR: 10.222.0.0/16
kubernetesVersion: "Automatic"
clusterDomain: "cluster.local"
defaultCRI: "ContainerdV2"
---
apiVersion: deckhouse.io/v1
kind: InitConfiguration
deckhouse:
  imagesRepo: registry.deckhouse.ru/deckhouse/ce
  registryDockerCfg: eyJhdXRocyI6IHsgInJlZ2lzdHJ5LmRlY2tob3VzZS5ydSI6IHt9fX0K
```



```
apiVersion: deckhouse.io/v1alpha1
kind: Cluster Configuration
type: Cloud
...
apiVersion: deckhouse.io/v1alpha1
kind: InitConfiguration
...
apiVersion: deckhouse.io/v1
kind: NodeGroup
...
apiVersion: deckhouse.io/v1
kind: IngressNginxController
```



```
apiVersion: deckhouse.io/v1
kind: YandexClusterConfiguration
layout: Standard
provider:
  cloudID: *!CHANGE_CloudID*
  folderID: *!CHANGE_FolderID*
  serviceAccountJSON: |
    {
      "id": "...",
      "service_account_id": "...",
      "created_at": "2022-08-04T05:38:34.756137618Z",
      "key_algorithm": "RSA_2048",
      "public_key": "-----BEGIN PUBLIC KEY-----...-----END PUBLIC KEY-----\n",
      "private_key": "-----BEGIN PRIVATE KEY-----...-----END PRIVATE KEY-----\n"
    }
masterNodeGroup:
  replicas: 1
  instanceClass:
    cores: 4
    memory: 8192
    imageID: fd83ica41cade1mj35sr # Стандартный образ ОС
    externalIPAddresses:
      - "Auto"
nodeNetworkCIDR: "10.241.32.0/20"
sshPublicKey: *!CHANGE_SSH_KEY*
```



```
apiVersion: deckhouse.io/v1
kind: YandexInstanceClass
metadata:
  name: worker
spec:
  cores: 4
  memory: 8192
  diskSizeGB: 30
---
apiVersion: deckhouse.io/v1
kind: NodeGroup
metadata:
  name: worker
spec:
  cloudInstances:
    classReference:
      kind: YandexInstanceClass
      name: worker
    maxPerZone: 1
    minPerZone: 1
    zones:
      - ru-central1-a
  disruptions:
    approvalMode: Automatic
  nodeType: CloudEphemeral
```

```
apiVersion: deckhouse.io/v1alpha1
kind: ModuleConfig
metadata:
  name: deckhouse
spec:
  version: 1
  enabled: true
  settings:
    releaseChannel: EarlyAccess
---
apiVersion: deckhouse.io/v1alpha1
kind: ModuleConfig
metadata:
  name: global
spec:
  version: 2
  settings:
    modules:
      publicDomainTemplate: "%s.example.com"
```


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Запустите установщик на персональном компьютере.

Linux / macOSWindows

i

Если при включенном VPN контейнер с установщиком не может получить доступ к сети, воспользуйтесь [инструкцией](#).

1

```
docker run --pull=always -it -v "$PWD/config.yml:/config.yml" -v "$HOME/.ssh:/tmp/.ssh/" -v "$PWD/dhctl-tmp:/tmp/dhctl" registry.deckhouse.ru/deckhouse/ce/install:early-access bash
```

Внутри контейнера выполните команду:

1

```
dhctl bootstrap --ssh-user=ubuntu --ssh-agent-private-keys=/tmp/.ssh/id_ed25519 --config=/config.yml
```

В параметре `--ssh-user` укажите имя пользователя по умолчанию для выбранного образа виртуальной машины. Для предложенного в руководстве образа это `ubuntu`.

Если установка была прервана...

<https://deckhouse.ru/products/kubernetes-platform/gs/yandex/step4.html>



**A FEW
MOMENTS LATER**


```
cluster — kovalkov@archmgmt: ~ — docker run --pull=always -it -v ~/cluster/config.yml:/config.yml -v ~/.ssh:/tmp/.ssh -v ~/cluster/dhctl-tmp:/tmp/dhctl registry.deckhouse.ru/d...

[ Create deckhouse.io/v1, Kind=IngressNginxController resources
  Manifest for IngressNginxController nginx
  🎉 Succeeded!
  Create deckhouse.io/v1, Kind=IngressNginxController resources (0.03 seconds)

[ Create deckhouse.io/v1, Kind=ClusterAuthorizationRule resources
  Manifest for ClusterAuthorizationRule admin
  🎉 Succeeded!
  Create deckhouse.io/v1, Kind=ClusterAuthorizationRule resources (0.03 seconds)

Checking if resource deckhouse.io/v1, Kind=IngressNginxController 'nginx' is ready...
Resource deckhouse.io/v1, Kind=IngressNginxController 'nginx' is ready!

Checking if resource deckhouse.io/v1, Kind=ClusterAuthorizationRule 'admin' is ready...
Resource deckhouse.io/v1, Kind=ClusterAuthorizationRule 'admin' is ready!

🔥 ~ Bootstrap: Create Resources (330.17 seconds)

🔥 ~ Bootstrap: Run post bootstrap actions
[ Set release channel to deckhouse module config
  🎉 Succeeded!
  Set release channel to deckhouse module config (0.12 seconds)
🔥 ~ Bootstrap: Run post bootstrap actions (0.12 seconds)

🔥 ~ Bootstrap: Clear cache
! ~ Next run of "dhctl bootstrap" will create a new Kubernetes cluster.
🔥 ~ Bootstrap: Clear cache (0.01 seconds)

🎉 Deckhouse cluster was created successfully!
🔴 ~ Common: Kubernetes Master Node addresses for SSH
archtest-master-0 | ssh ubuntu@89.169.138.233
🔴 ~ Common: Kubernetes Master Node addresses for SSH (0.00 seconds)

[deckhouse] root@4777dfe05c0c / #
```



```
[ubuntu@archtest-master-0:~$ sudo -i d8 k get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
archtest-master-0	Ready	control-plane,master	28m	v1.32.10
archtest-worker-6dbfa292-56989-z67j9	Ready	worker	16m	v1.32.10

```
[ubuntu@archtest-master-0:~$ sudo -i d8 k get pods --all-namespaces
```

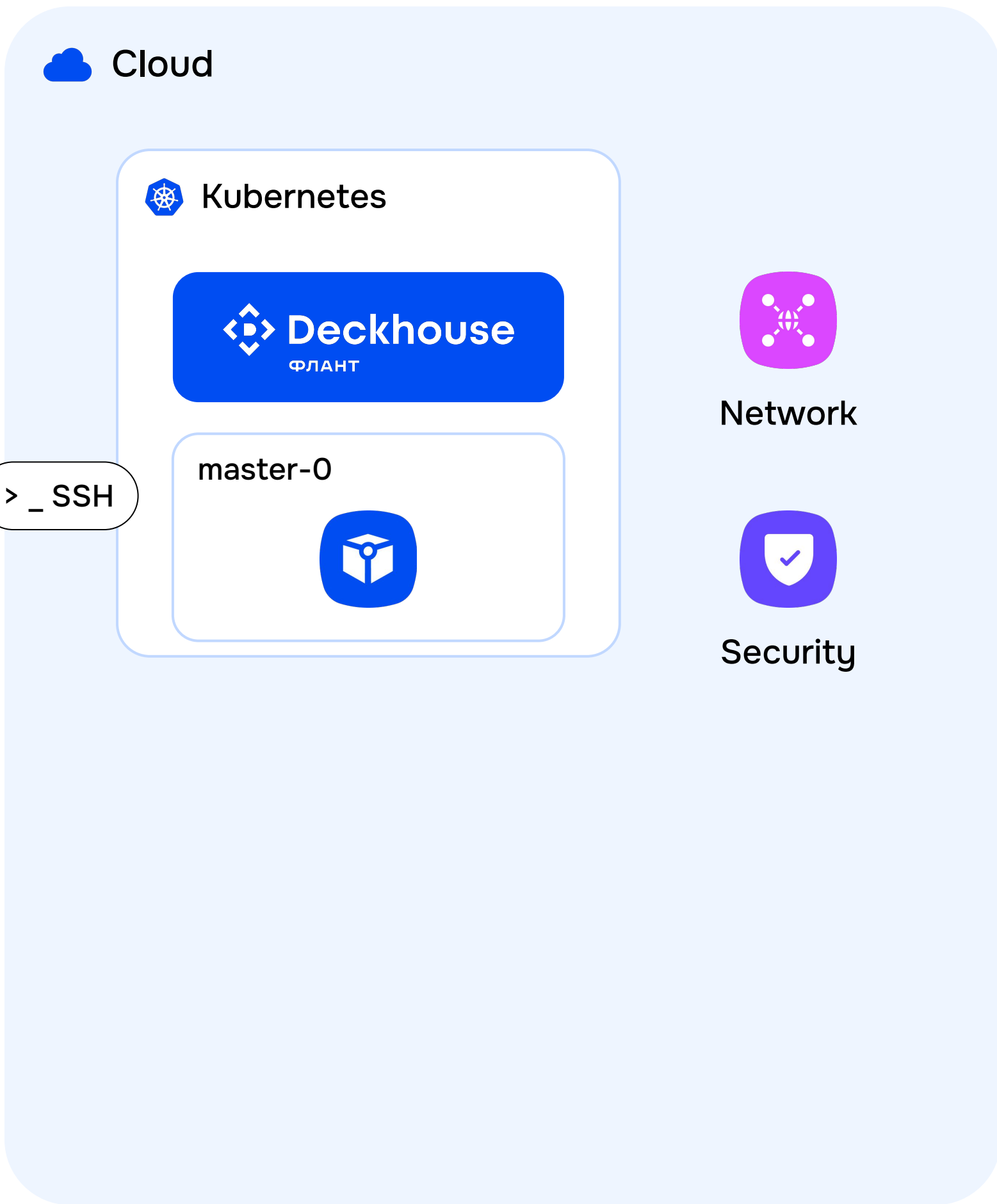
NAMESPACE	NAME	READY	STATUS	RESTARTS	AGE
d8-admission-policy-engine	gatekeeper-audit-5445487c64-bg9x9	3/3	Running	5 (11m ago)	13m
d8-admission-policy-engine	gatekeeper-controller-manager-866f7b8795-qm9hs	2/2	Running	2 (11m ago)	13m
d8-cert-manager	cert-manager-74ddb46ccd-fcv5s	2/2	Running	2 (4m28s ago)	13m
d8-cert-manager	webhook-54f9d4c4d7-5xkhj	1/1	Running	0	13m
d8-chrony	chrony-fpgdg	3/3	Running	0	13m
d8-chrony	chrony-master-jqklc	3/3	Running	0	13m
d8-cloud-instance-manager	bashible-apiserver-5c55987bd9-mn162	1/1	Running	0	14m
d8-cloud-instance-manager	early-oom-fwgcr	2/2	Running	0	46s
d8-cloud-instance-manager	early-oom-s2r4v	2/2	Running	0	78s
d8-cloud-instance-manager	machine-controller-manager-fc797f966-c4svw	2/2	Running	2 (5m3s ago)	14m
d8-cloud-instance-manager	registry-packages-proxy-544bf777c7-5s29b	2/2	Running	0	14m
d8-cloud-provider-yandex	cloud-controller-manager-646ddc9d85-bwmdh	1/1	Running	5 (5m4s ago)	15m
d8-cloud-provider-yandex	cloud-data-discoverer-6bcbd7bcf4-5fhxs	2/2	Running	0	15m
d8-cloud-provider-yandex	csi-controller-848546cc98-lsqkn	5/5	Running	8 (4m51s ago)	15m
d8-cloud-provider-yandex	csi-node-8sccm	2/2	Running	0	48s
d8-cloud-provider-yandex	csi-node-n9vx7	2/2	Running	0	74s
d8-cni-simple-bridge	simple-bridge-f7r26	1/1	Running	0	29s
d8-cni-simple-bridge	simple-bridge-lq74t	1/1	Running	0	63s
d8-console	backend-67c54b5c55-gtqdj	1/1	Running	0	13m
d8-console	console-dex-authenticator-69c84d7694-gxdsc	2/2	Running	0	5m31s
d8-console	frontend-7d4b8765bd-8z4f9	1/1	Running	0	13m
d8-console	observability-gw-68b6ccf86c-s88cf	1/1	Running	0	13m
d8-dashboard	api-77f6d64ff6-df5f9	2/2	Running	0	13m
d8-dashboard	auth-78d647df9f-f9rpx	2/2	Running	0	13m
d8-dashboard	dashboard-dex-authenticator-b448d5d97-q9lq7	2/2	Running	0	5m31s
d8-dashboard	metrics-scraper-5ccc7ccdcc-xkh7g	1/1	Running	0	13m
d8-dashboard	web-78ffb97c46-vcq9n	2/2	Running	0	13m
d8-ingress-nginx	controller-nginx-b4hwz	3/3	Running	0	11m

d8-snapshot-controller	webhooks-77ccbf85-ltxcv	1/1	Running	0		13m
d8-system	deckhouse-5bcb687964-chls8	2/2	Running	2	(4m27s ago)	16m
d8-system	deckhouse-tools-769c895dc-l7gz5	1/1	Running	0		13m
d8-system	deckhouse-tools-dex-authenticator-bcf89f64d-hhm5c	2/2	Running	0		5m47s
d8-system	documentation-8559ddffd-mn8kb	3/3	Running	3	(3m54s ago)	13m
d8-system	documentation-dex-authenticator-579db9dbc9-snbqv	2/2	Running	0		5m47s
d8-system	terraform-auto-converger-6ffd97ff9b-n4lq2	2/2	Running	0		14m
d8-system	terraform-state-exporter-779869b84d-lp7sx	2/2	Running	0		14m
d8-system	webhook-handler-75d7569947-vddcv	1/1	Running	0		15m
d8-upmeter	smoke-mini-a-0	1/1	Running	0		109s
d8-upmeter	smoke-mini-b-0	1/1	Running	0		50s
d8-upmeter	status-7fcfc9656d-5fzxt	1/1	Running	0		13m
d8-upmeter	status-dex-authenticator-89cd5bb49-lm84	2/2	Running	0		5m47s
d8-upmeter	upmeter-0	2/2	Running	0		4m12s
d8-upmeter	upmeter-agent-qx692	1/1	Running	0		5m16s
d8-upmeter	upmeter-dex-authenticator-5fd96bd6bd-lttw2	2/2	Running	0		5m46s
d8-upmeter	webui-849dcc99dc-dvzq5	1/1	Running	0		13m
d8-user-authn	dex-74d58bc859-k58rr	2/2	Running	0		13m
d8-user-authn	kubeconfig-generator-74f96ff66d-l7zkn	1/1	Running	0		5m46s
kube-system	d8-control-plane-manager-lhnz5	6/6	Running	0		5m24s
kube-system	d8-kube-dns-67f4787d4f-lqf57	2/2	Running	0		45s
kube-system	d8-kube-dns-67f4787d4f-r26jp	2/2	Running	0		78s
kube-system	d8-kube-proxy-nx5h7	2/2	Running	0		74s
kube-system	d8-kube-proxy-pd9l5	2/2	Running	0		43s
kube-system	etcd-archtest-master-0	1/1	Running	0		22m
kube-system	kube-apiserver-archtest-master-0	2/2	Running	0		4m30s
kube-system	kube-controller-manager-archtest-master-0	1/1	Running	2	(5m5s ago)	21m
kube-system	kube-scheduler-archtest-master-0	1/1	Running	2	(5m4s ago)	21m
kube-system	kubernetes-api-proxy-archtest-master-0	2/2	Running	0		27m
kube-system	kubernetes-api-proxy-archtest-worker-6dbfa292-56989-z67j9	2/2	Running	0		16m
kube-system	vpa-admission-controller-75955cd845-8gjsn	2/2	Running	0		2m55s
kube-system	vpa-recommender-78f84c64f5-4p7zc	2/2	Running	0		2m55s
kube-system	vpa-updater-667cf56c7-t4v8q	2/2	Running	0		2m55s

ubuntu@archtest-master-0:~\$


```
$ docker run --pull=always -it -v "$PWD/config.yml:/config.yml" -v  
"$HOME/.ssh:/tmp/.ssh/" -v "$PWD/dhctl-tmp:/tmp/dhctl"  
registry.deckhouse.ru/deckhouse/ce/install:stable dhctl bootstrap  
--ssh-user=ubuntu --ssh-agent-private-keys=/tmp/.ssh/id_rsa  
--config=/config.yml
```

> _SSH

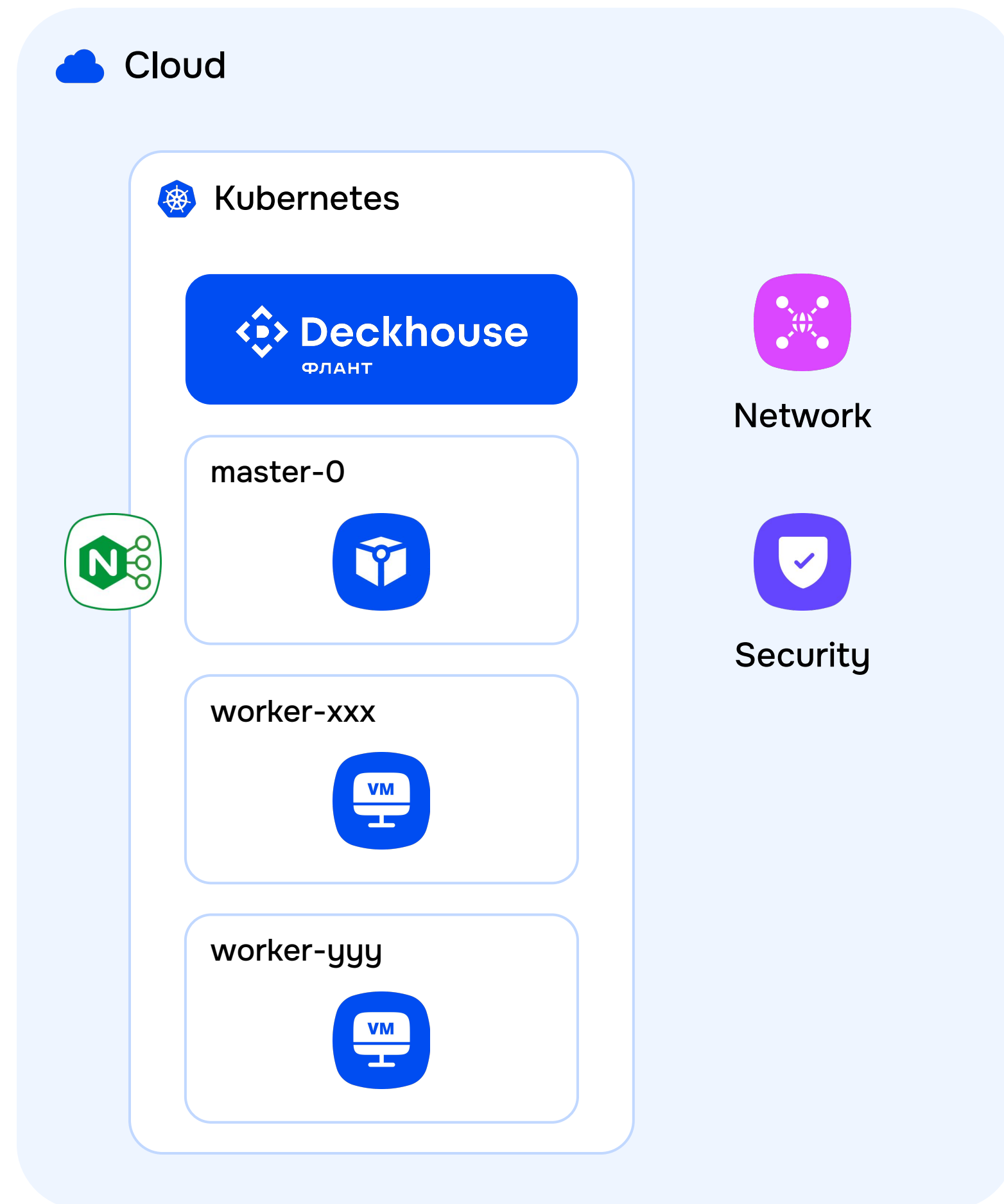


Базовый кластер готов





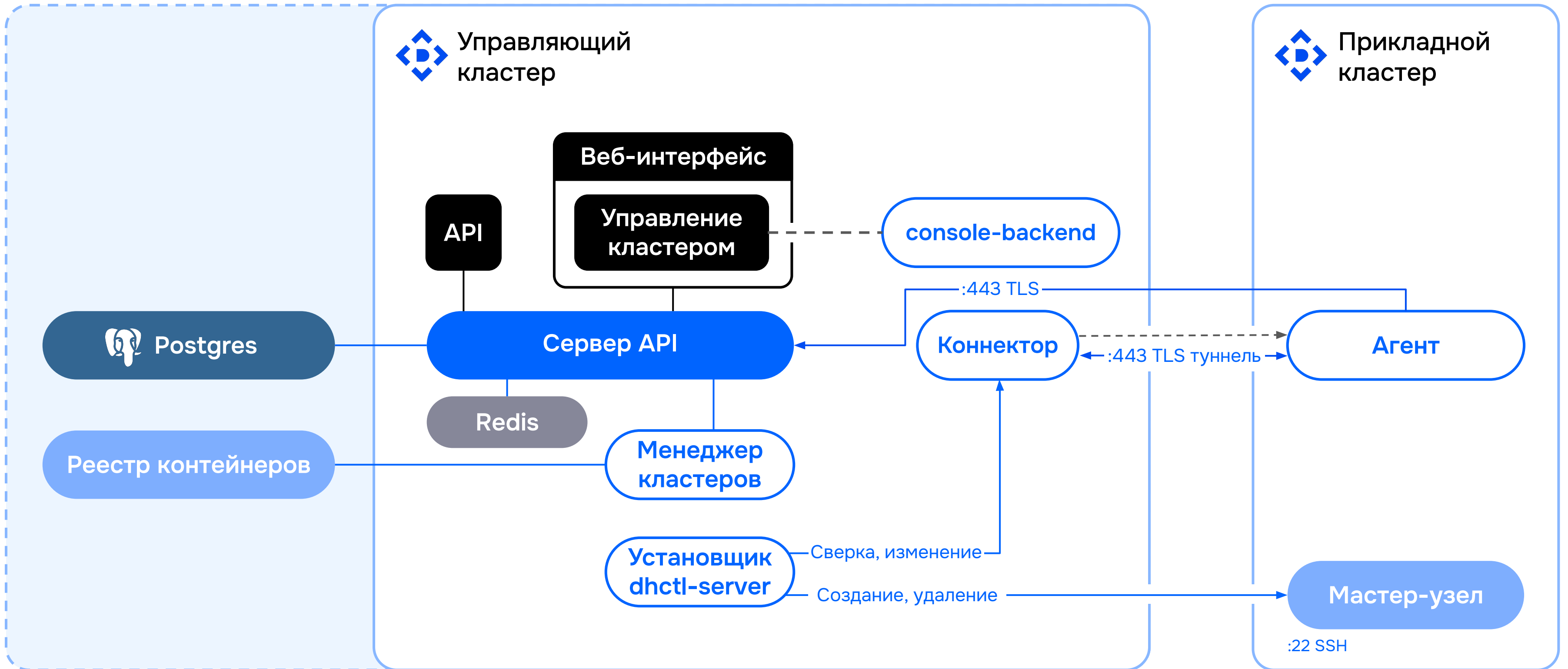
Кластер готов к работе!



Итого

Время РоС: ~1 вечер





Deckhouse Commander

Private browsing

commander.archdev.deckhouse-presales.hf.flant.com/workspaces/vzwxr/cluster-templates/b09274d8-1fde-4741

Sign in

Deckhouse Commander

Главное

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Cluster Templates

Inventory

Audit

Archive

RBAC

DK

kovalkov - Yandex Cloud

v2

Create cluster >

Template Parameters

Versions (2)

Clusters (1)

Change History

Input parameters

INFRASTRUCTURE

Kubernetes

Placement

SSH Parameters

Add group +

Kubernetes *

In this section, you must specify the [ClusterConfiguration](#). This is the main configuration file that defines the global settings for the Kubernetes cluster, such as the Kubernetes version, network settings, API server parameters, etc.

```
4  apiVersion: deckhouse.io/v1
5  kind: ClusterConfiguration
6  clusterType: Cloud
7  cloud:
8    provider: Yandex
9    # Префикс объектов, создаваемых в облаке при установке.
10   prefix: {{ .prefix | quote }}
11   # Адресное пространство Pod'ов кластера.
```




**A FEW
MOMENTS LATER**

Deckhouse Commander

Private browsing

commander.archdev.deckhouse-presales.hf.flant.com/workspaces/vzwxr/clusters/active?sort=field=created_at%

Deckhouse Commander

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Clusters

Create cluster +

Attach

Creation date (newest first)

Clu...

Template

Status

Deckhouse version

Kubernetes version

X

Found 1 clusters

Total 1 clusters

archtest

READY

CONFIGURED

Template: [kovalkov - Yandex Cloud](#)

v2

Update date: 2026.04.07, 03:56:20

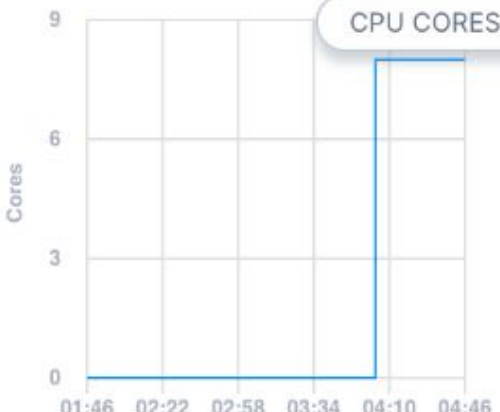
Warnings: 2

Availability: 100%

Deckhouse: v1.74.15 EarlyAccess

Kubernetes: v1.32.10

CPU CORES



Time	Cores
01:46	0
02:22	0
02:58	0
03:34	0
04:10	9
04:46	9

MEMORY



Time	Memory (Gigabytes)
01:46	0
02:22	0
02:58	0
03:34	0
04:10	16
04:46	16

DISKS



Time	Disks (Gigabytes)
01:46	0
02:22	0
02:58	0
03:34	0
04:10	180
04:46	180

NODES



Time	Nodes
01:46	0
02:22	0
02:58	0
03:34	0
04:10	2
04:46	2

System / System Management / X

:warning: migrate to Cluster API X

+

Private browsing

commander.archdev.deckhouse-presales.hf.flant.com/workspaces/vzwxr/clusters/4ebcfbc8-2f39-46a6-b697-c4

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+ YAML

System / System Management / Deckhouse

Deckhouse EE

v1.74.15 EarlyAccess

Kubernetes

v1.32.10

Tools


```
cluster — ubuntu@archtest-master-0: ~ — -bash — 134x34

[04:48:49]: ~/cluster $ d8 k -n d8-commander get deployments
NAME                                     READY    UP-TO-DATE    AVAILABLE    AGE
ampg-connector-e9e653d8-47a2-4e33-b10c-1d77ec99c4b7 1/1      1             1            63m
backend                                     2/2      2             2            13d
cable                                     2/2      2             2            13d
cluster-checker                           2/2      2             2            13d
cluster-manager                           2/2      2             2            13d
cluster-task-checker                     2/2      2             2            13d
commander-dex-authenticator               2/2      2             2            13d
console-backend                           2/2      2             2            13d
console-frontend                           2/2      2             2            13d
dhctl-57d69e1f-1-74-15                    1/1      1             1            11d
dhctl-57d69e1f-1-75-3                    1/1      1             1            16h
frontend                                  2/2      2             2            13d
redis                                     1/1      1             1            13d
sidekiq                                   2/2      2             2            13d
[04:48:51]: ~/cluster $
```



```
cluster — ubuntu@cmdr-kovalkov-master-0: ~ — ssh -i .ssh/id_rsa ubuntu@89.169.141.201 — 134x34

[ubuntu@cmdr-kovalkov-master-0:~$ sudo -i d8 k -n d8-commander-agent get pods
NAME                                READY    STATUS    RESTARTS   AGE
commander-agent-9984fff48-p25w7    1/1      Running   0           58m
ubuntu@cmdr-kovalkov-master-0:~$
```



БОНУС

Deckhouse Commander

GitHub - unreturned/yc-d8-airg

+

Private browsing

github.com/unreturned/yc-d8-airgapped

Sign in

README

MIT license

Yandex Cloud Isolated Environment with Deckhouse Kubernetes

Terraform/OpenTofu configuration for deploying isolated infrastructure in Yandex Cloud with HTTP proxy for Kubernetes.

License MIT

What is this?

This project provisions a ready-to-use infrastructure for **single-node Deckhouse Kubernetes Platform** in an isolated Yandex Cloud environment with **automatic Deckhouse installation**:

- **Bastion** — public host with HTTP proxy (Tinyproxy), Nginx reverse proxy with Let's Encrypt, and NTP server
- **Registry** — private host with Harbor Container Registry, NFS server, and **Deckhouse installer**
- **Master** — Kubernetes master node with a separate disk for etcd

Key feature: Deckhouse is installed **automatically** during `terraform apply` via cloud-init on the registry host.



<https://github.com/unreturned/yc-d8-airgapped>


```
[16:46:10]: ~/yc-d8-airgapped (main)$ make init >/dev/null && make apply >/dev/null
```

```
[16:46:26]: ~/yc-d8-airgapped (main)$ make urls
```

```
[INFO] Deckhouse UI & Harbor Access Information (sslip.io)
```

```
[INFO] Bastion IP: 111.88.253.121
```

```
[INFO] sslip.io domain: 111-88-253-121.sslip.io
```

```
[INFO] Deckhouse URLs (with valid SSL after Deckhouse installation):
```

```
  Console:      https://console.111-88-253-121.sslip.io
```

```
[INFO] Deckhouse Login credentials:
```

```
  Email:      admin@deckhouse.io
```

```
  Password: 7md03yxgzt
```

```
[INFO] Harbor Registry:
```

```
  URL:      https://harbor.111-88-253-121.sslip.io
```

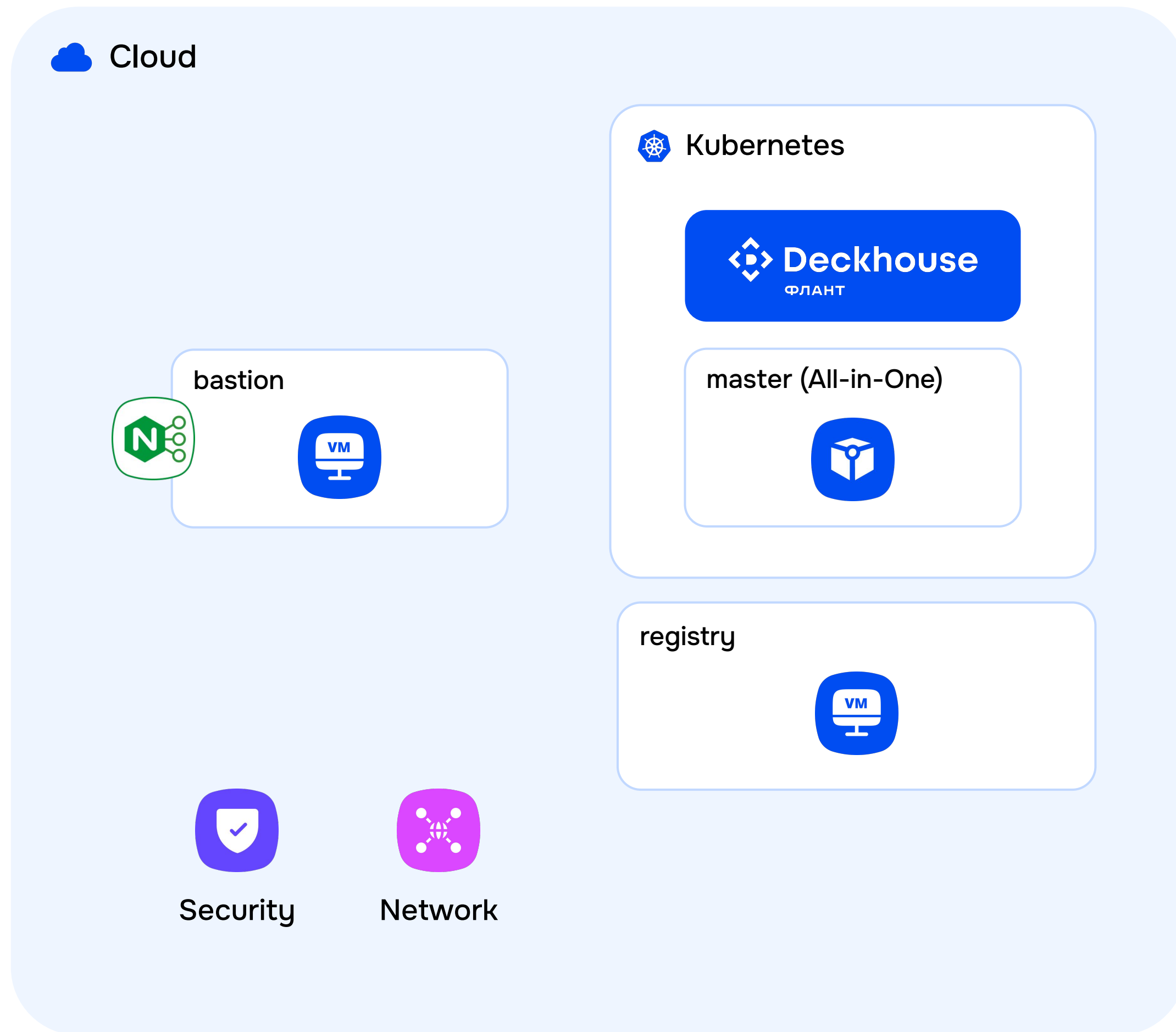
```
  Login:    admin
```

```
  Password: Harbor12345
```

```
[16:46:30]: ~/yc-d8-airgapped (main)$
```




Лаба готова к работе!



Спасибо за внимание!

 @deckhouse_news

 github.com/deckhouse

